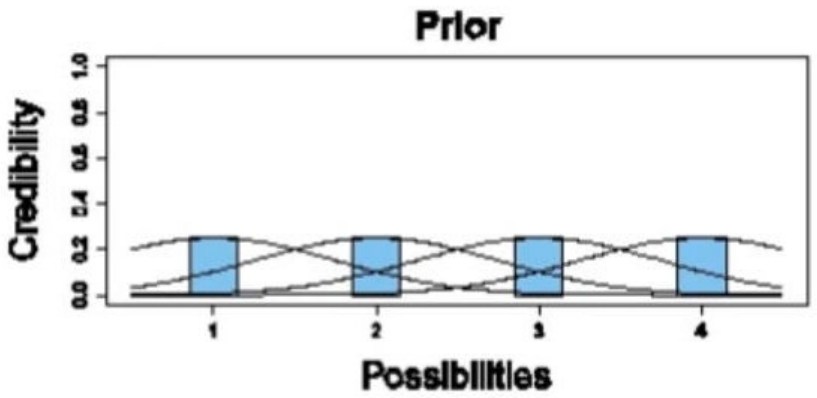
Inferencia bayesiana con base en la distribución posterior

Dr. Héctor Nájera
Dr. Curtis Huffman

19/02/2026

Análisis bayesiano de datos

- El papel de los datos es reubicar credibilidad entre posibilidades (i.e., aprender).
- Las posibilidades son valores de parámetros en un modelo probabilístico (μ, σ)
- Reubicamos credibilidad a valores de parámetros consistentes con los datos



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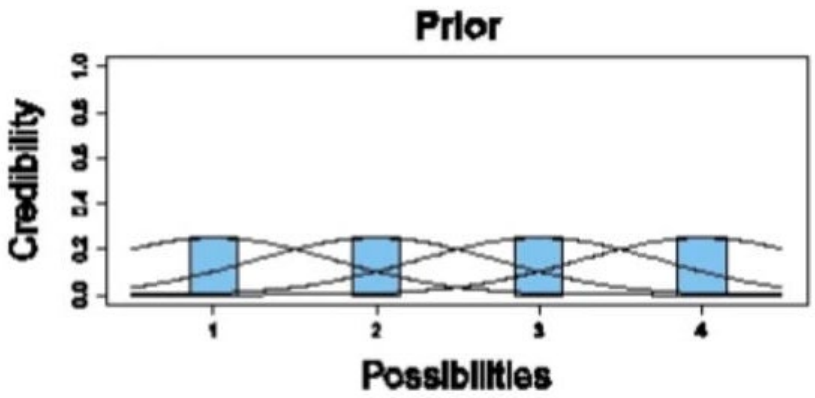
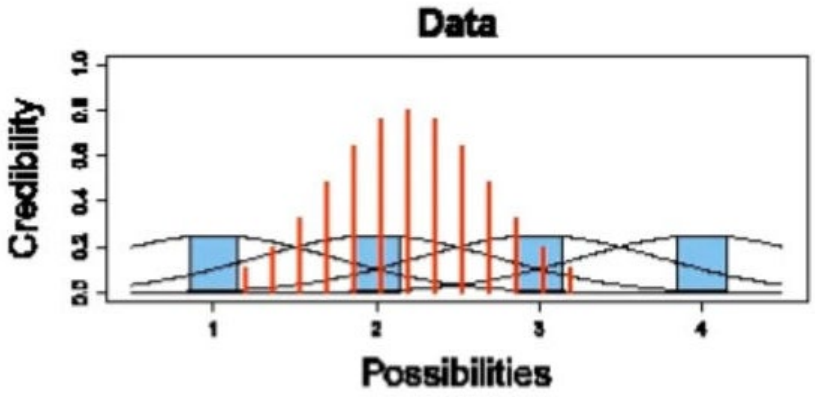




Cláse pasada:

Teorema de Bayes

Formula:

Likelihood distribution

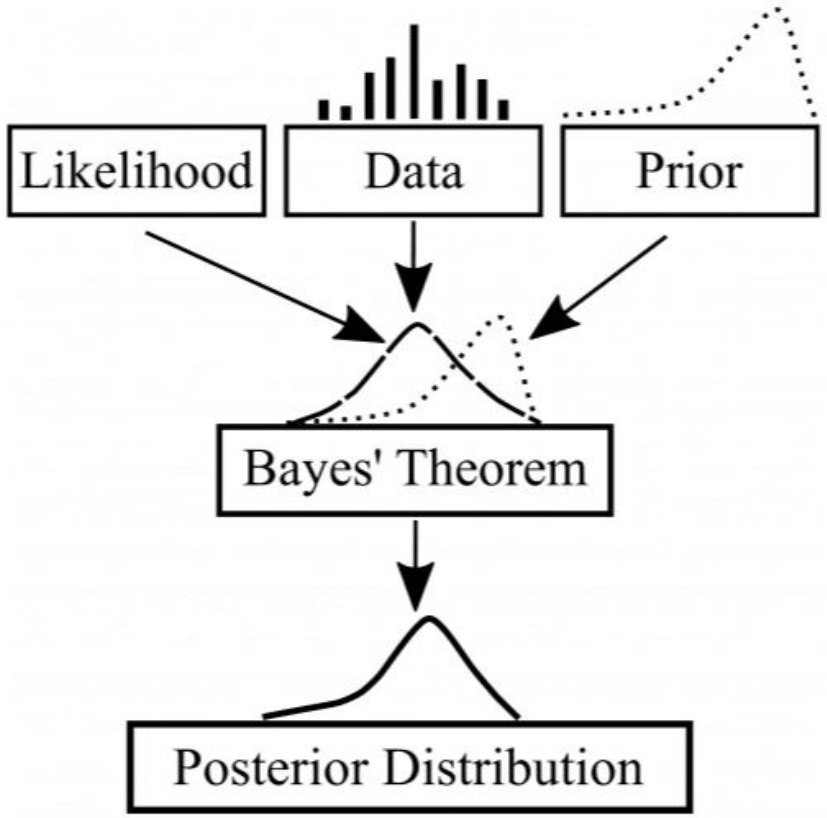
Prior distribution

$$f(\text{model} \mid \text{data}) = \frac{f(\text{data} \mid \text{model}) \times f(\text{model})}{f(\text{data})}$$

Bayesian Model

Data

Bayesian Model:



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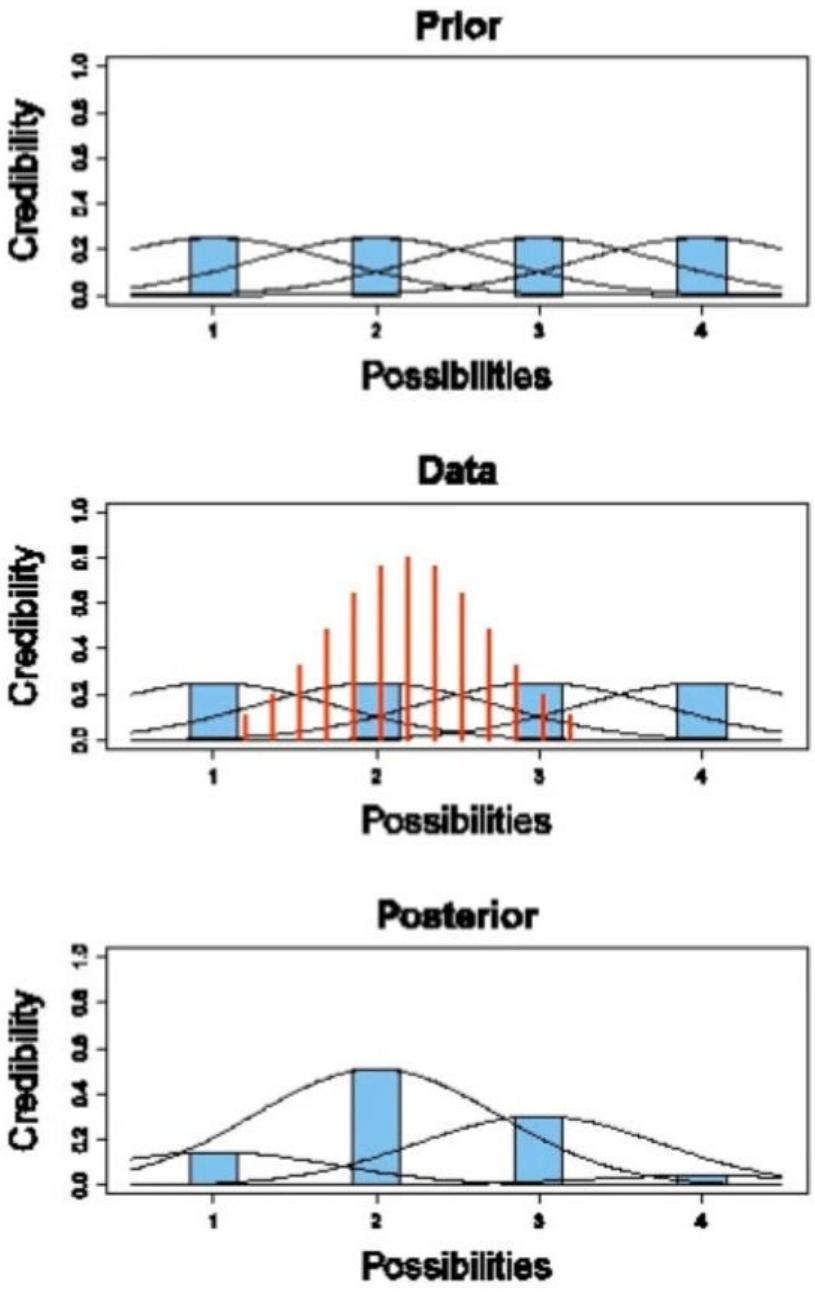
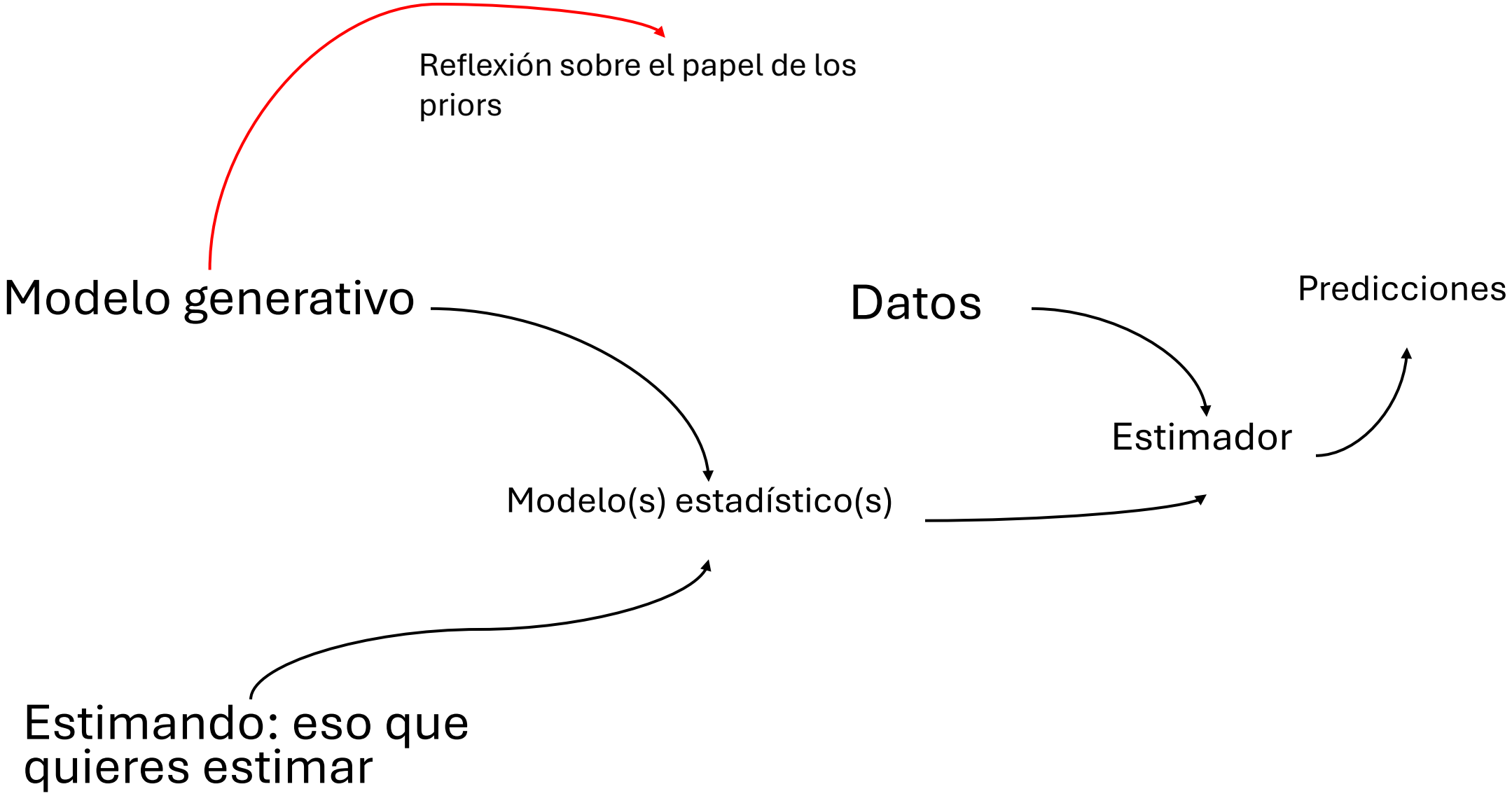



Reflexión sobre el papel de los priors
Modelo generativo
Datos
Predicciones
Estimador
Modelo(s) estadístico(s)
Estimando: eso que quieres estimar

The prior can generally only be understood in the context of the likelihood*

Andrew Gelman† Daniel Simpson‡ Michael Betancourt§

28 Aug 2017

Abstract

A key sticking point of Bayesian analysis is the choice of prior distribution, and there is a vast literature on potential defaults including uniform priors, Jeffreys' priors, reference priors, maximum entropy priors, and weakly informative priors. These methods, however, often manifest a key conceptual tension in prior modeling: a model encoding true prior information should be chosen without reference to the model of the measurement process, but almost all common prior modeling techniques are implicitly motivated by a reference likelihood. In this paper we resolve this apparent paradox by placing the choice of prior into the context of the entire Bayesian analysis, from inference to prediction to model evaluation.

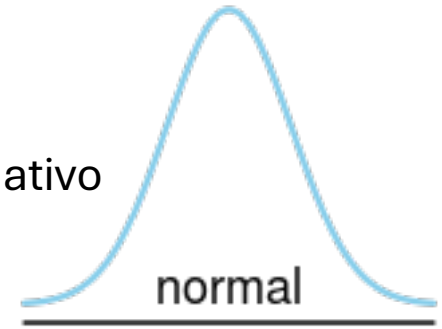
Priors en el razonamiento bayesiano

Prior / A priori

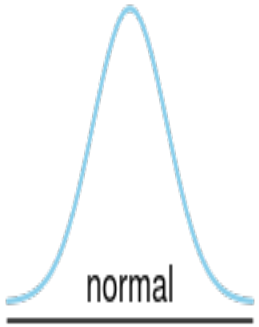
$p(\theta)$

No sé nada

Más informativo

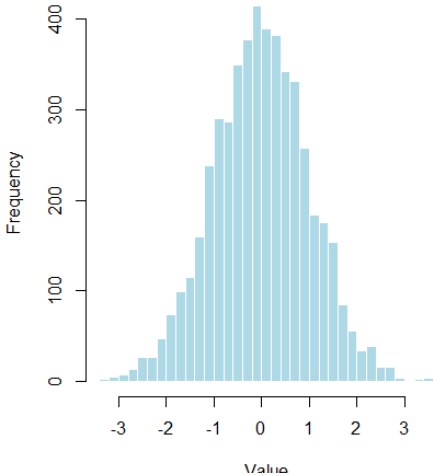


Mucho más informativo



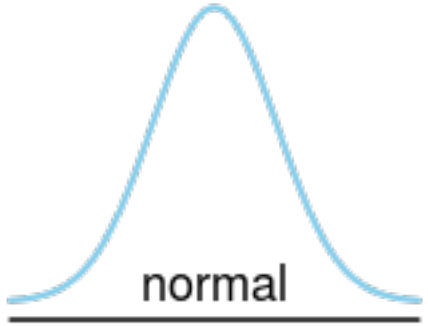
Verosimilitud

$P(D|\theta)$

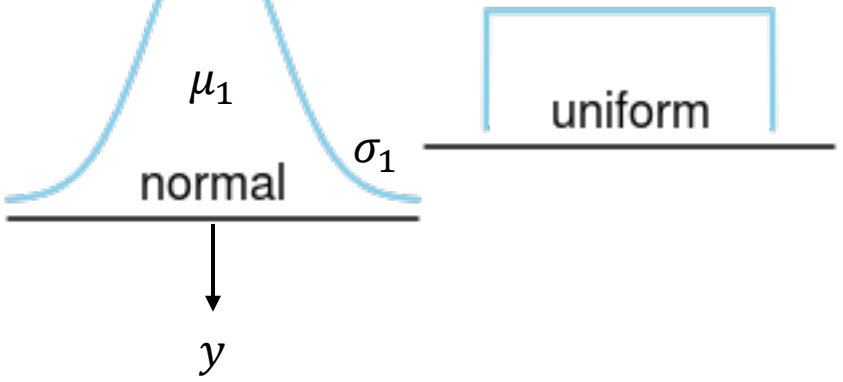


Posterior

$P(\theta|D)$



$$= \alpha + \beta_1 Edad + \beta_2 G$$

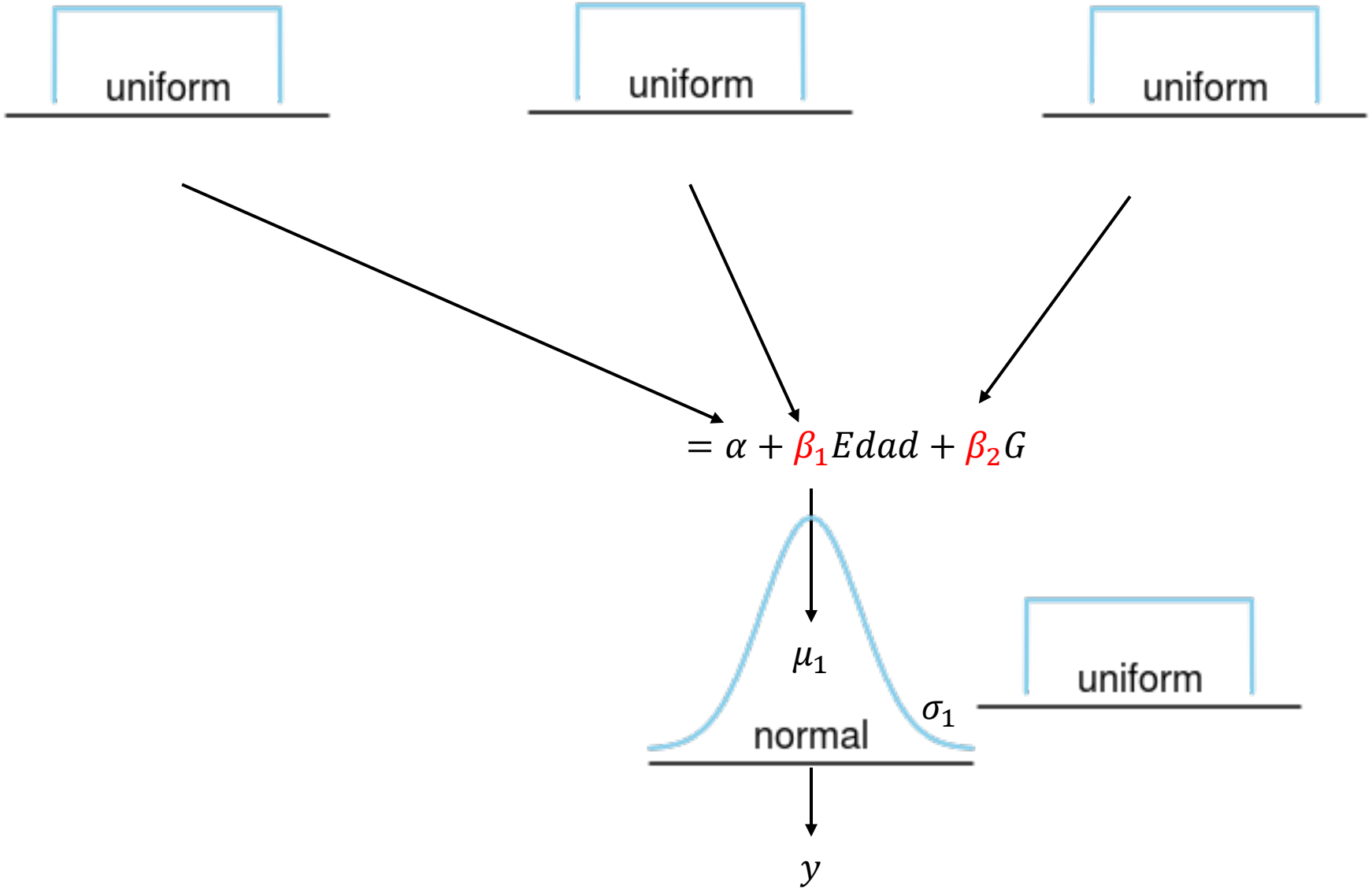


y = Salario por hora

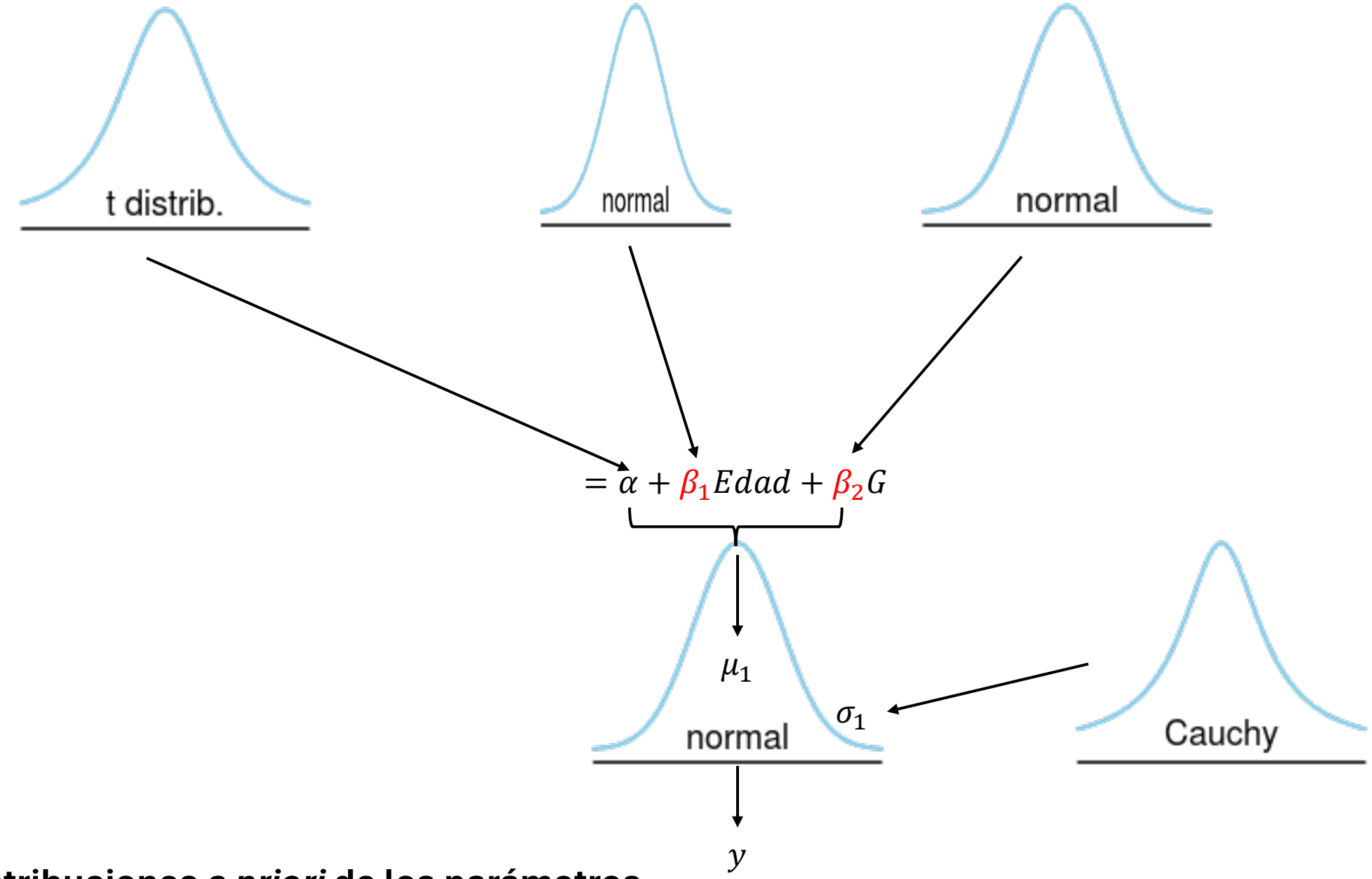
¿Qué priors elegirían?

Efecto lineal de la edad sobre el salario

Efecto de graduarse de cierto programa sobre el salario



Distribuciones a priori de los parámetros: El modelo "determinístico" es el mismo pero el modelo generativo no



Distribuciones *a priori* de los parámetros

Escritura del modelo bayesiano

$S_i \sim \text{Normal}(\mu_i, \sigma)$ [likelihood]

$\mu_i = \alpha + \beta_1 * B1_i + \beta_2 * B2_i$ [linear model]

$\mu \sim \text{Normal}(10000, 500)$ [μ prior]

$\sigma \sim \text{Exponential}(1)$ [σ prior]

$\alpha \sim \text{Normal}(0, 1000)$ [α prior]

$\beta_1 \sim \text{Normal}(0, 100)$ [β1 prior, edad]

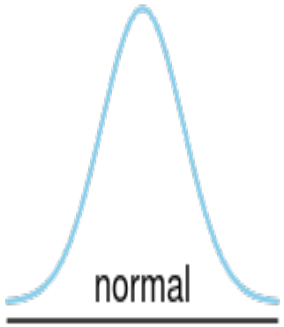
$\beta_2 \sim \text{Normal}(0, 500)$ [β2 prior, escuela de graduación]

¿Por qué los priors (a prioris) se eligen en el contexto de la verosimilitud?

La edad tiene ciertos rangos

Los salarios también

Efectos de escala de Y



$$= \alpha + \beta_1 Edad + \beta_2 G$$

Condiciones mínimas:

Regularizar inferencias en el contexto de la verosimilitud

Otro ejemplo

Beautiful British Parents Have More Daughters

Satoshi Kanazawa View all authors and affiliations

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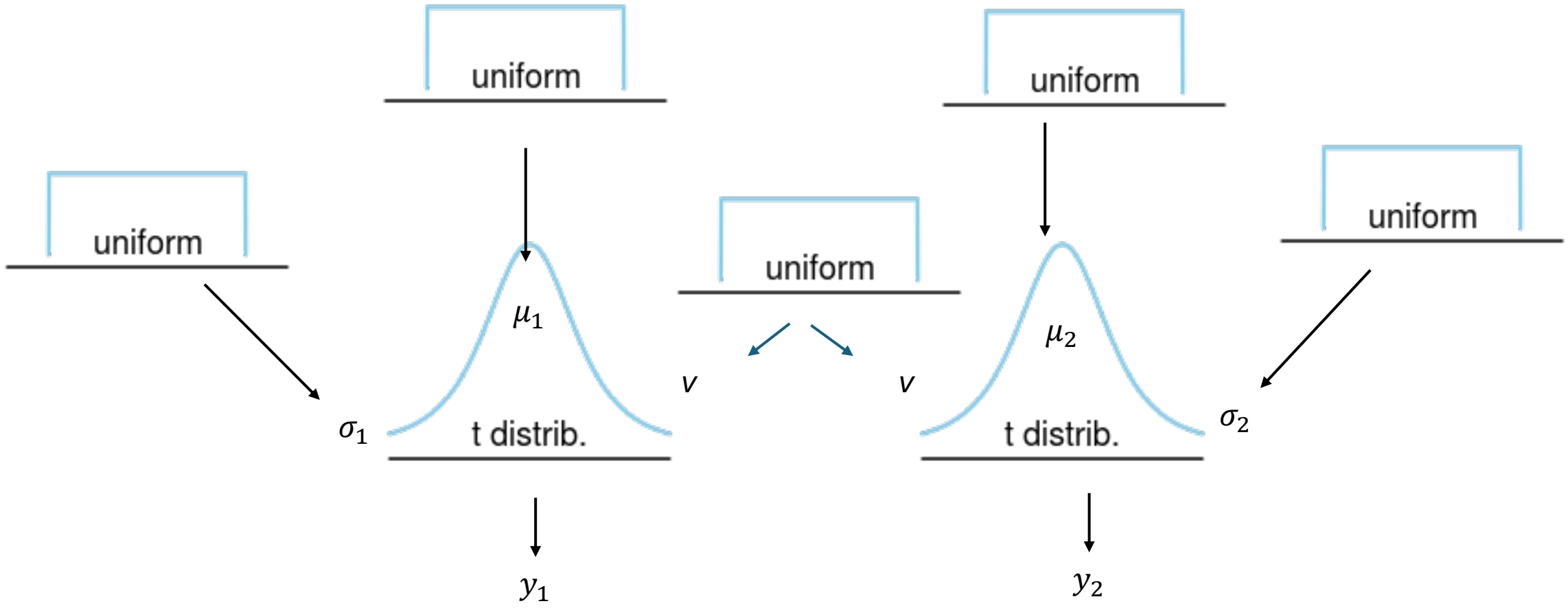
Abstract

The generalized Trivers-Willard hypothesis proposes that parents who possess any heritable trait that increases male reproductive success at a greater rate than female reproductive success in a given environment will have a higher-than-expected offspring sex ratio (more sons), and parents who possess any heritable trait that increases female reproductive success at a greater rate than male reproductive success in a given environment will have a lower-than-expected offspring sex ratio (more daughters). One heritable trait that increases the reproductive success of daughters much more than that of sons is physical attractiveness. The generalized Trivers-Willard hypothesis therefore predicts that physically attractive parents have more daughters. Further, if beautiful parents have more daughters and physical attractiveness is heritable, then over evolutionary history women on average should gradually become more attractive than men. The analysis of the prospectively longitudinal National Child Development Study in the United Kingdom replicates earlier findings with an American sample and confirms both hypotheses. British children who are rated by their teachers as "attractive" at age 7 have 23% higher odds of having a daughter 40 years later (proportion sons = 0.50127); those who are rated by their teachers as "unattractive" at age 7 have 25% higher odds of having a son 40 years later (proportion sons = 0.56285).

Y {"niña","niño"}

X {"No guapos", "Guapos"}

N=3000


uniform
uniform
uniform
uniform
uniform
μ_1
μ_2
v
v
σ_1
σ_2
t distrib.
t distrib.
y_1
y_2

La posterior va a ser proporcional a la verosimilitud

Prior / A priori

$p(\theta)$

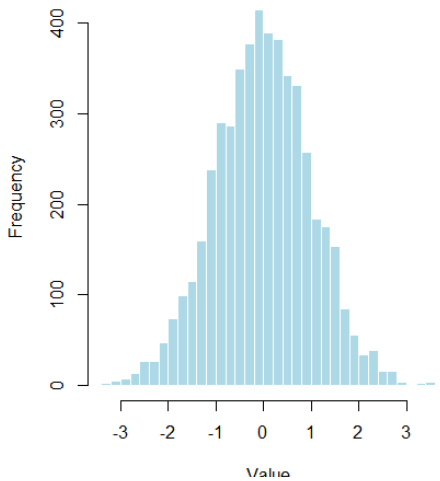
Verosimilitud

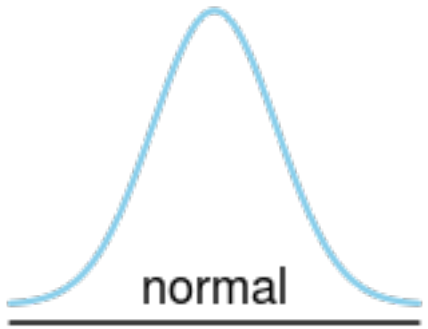
$P(D|\theta)$

Posterior

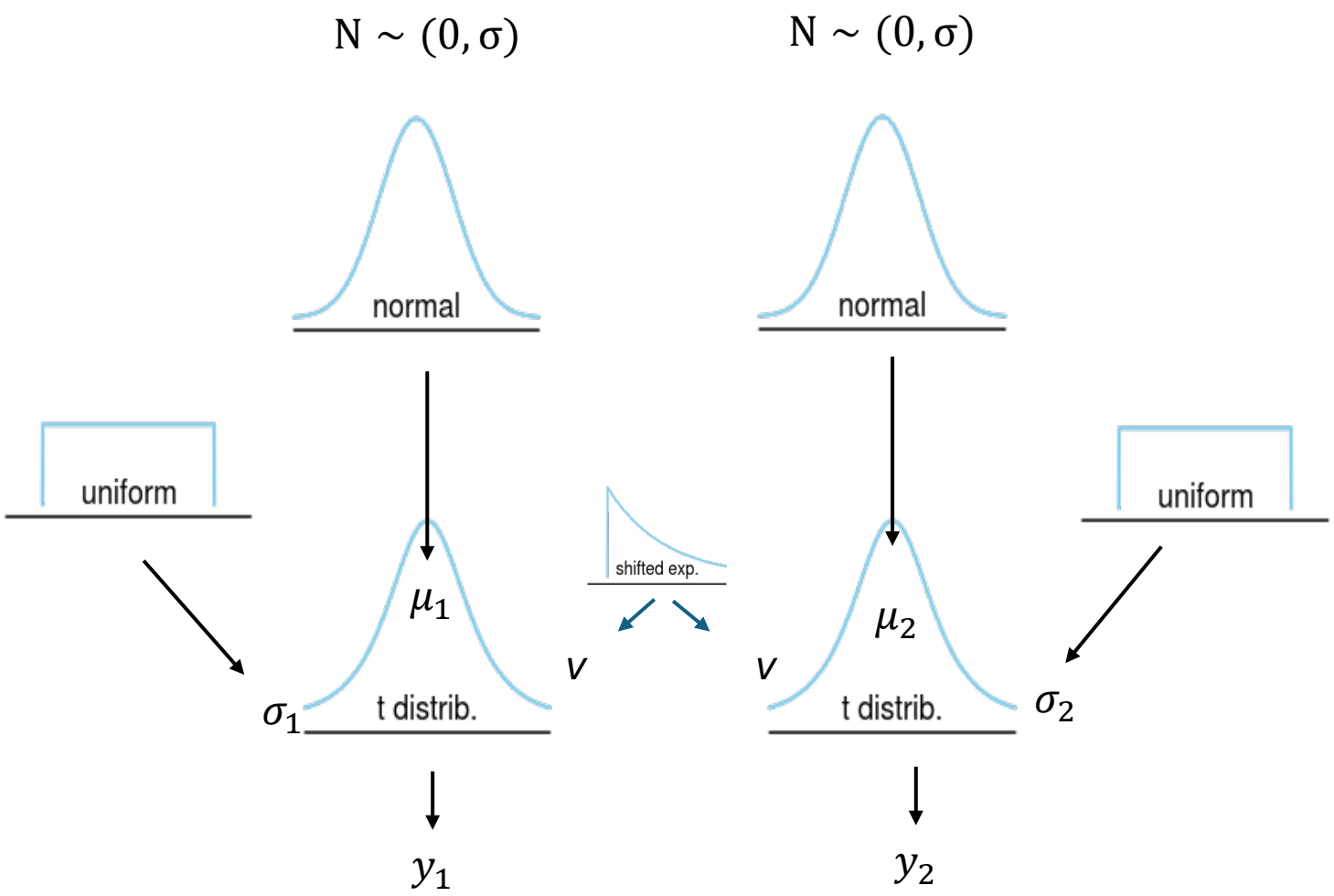
$P(\theta|D)$

No sé nada





El a priori uniforme supone que no sabemos nada de que la relación de nacimientos de niñas y niños es casi 50%.


N ~ (0, σ)
N ~ (0, σ)
normal
normal
uniform
uniform
shifted exp.
μ_1
μ_2
v
v
σ_1
σ_2
t distrib.
t distrib.
y_1
y_2

El prior es más que una distribución de probabilidad, piensa generativamente

Prior / A priori

$p(\theta)$

Verosimilitud

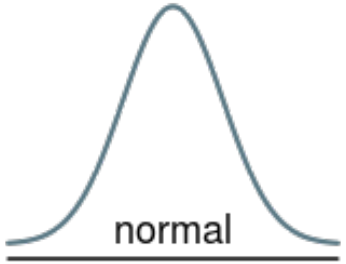
$P(D|\theta)$

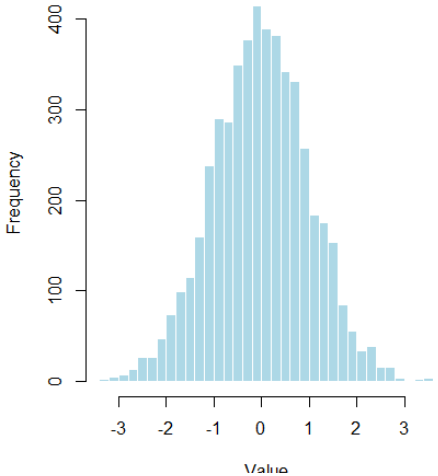
Posterior

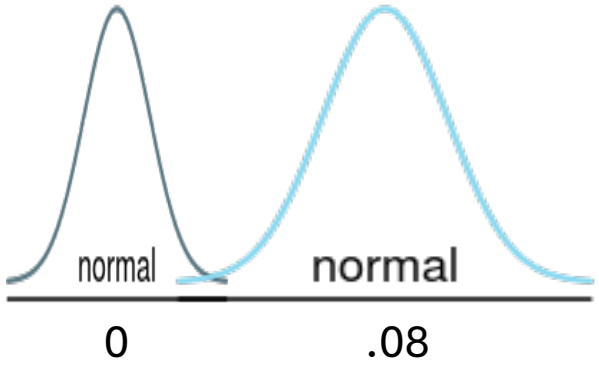
$P(\theta|D)$

No sé nada

50/50







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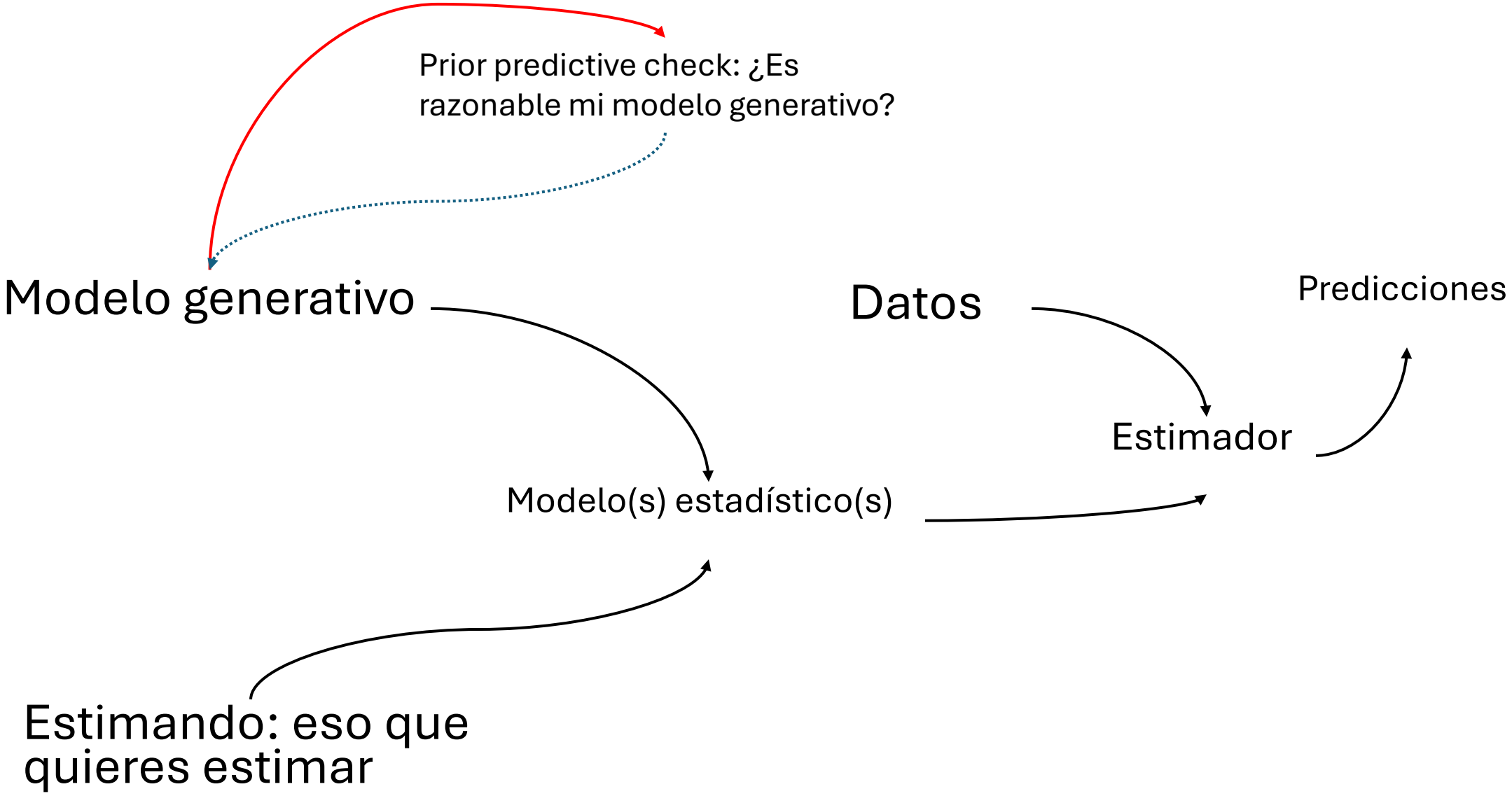
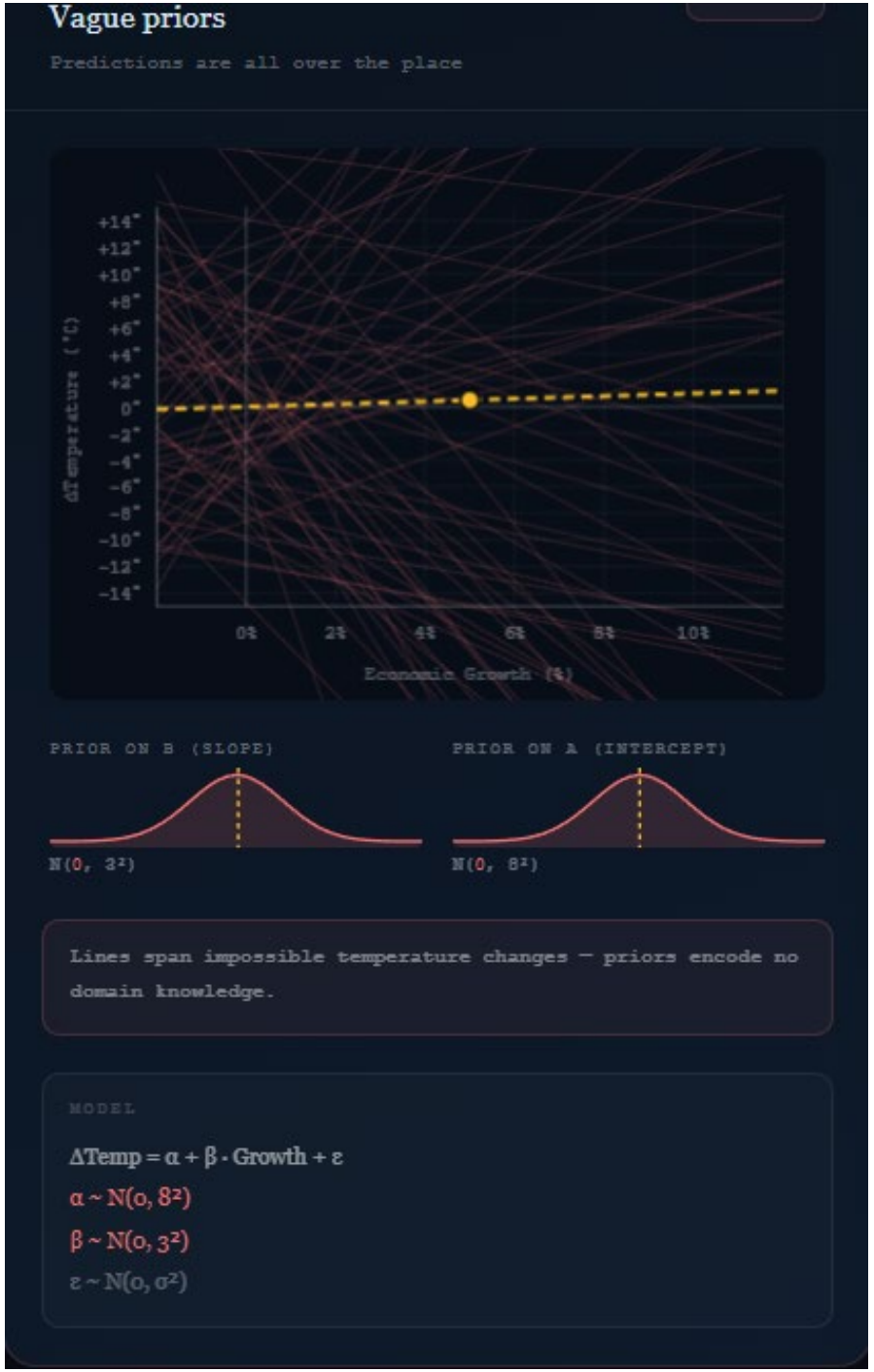
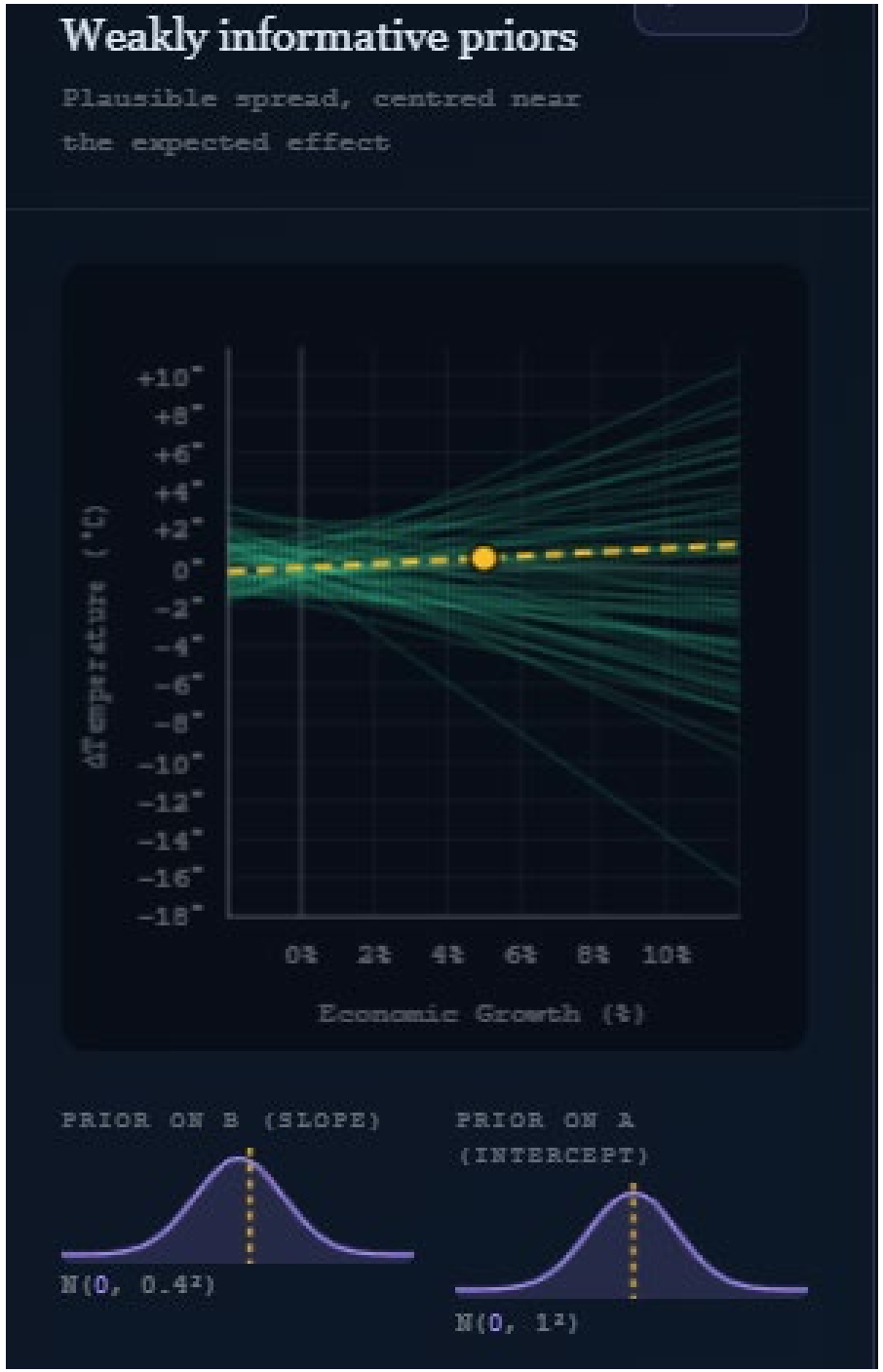

Prior predictive check: ¿Es razonable mi modelo generativo?
Modelo generativo
Datos
Predicciones
Estimador
Modelo(s) estadístico(s)
Estimando: eso que quieres estimar

Prior predictive checks

¿Pueden tus priors, bajo el modelo generativo, producir los valores esperados (e inesperados) con cierta probabilidad)?

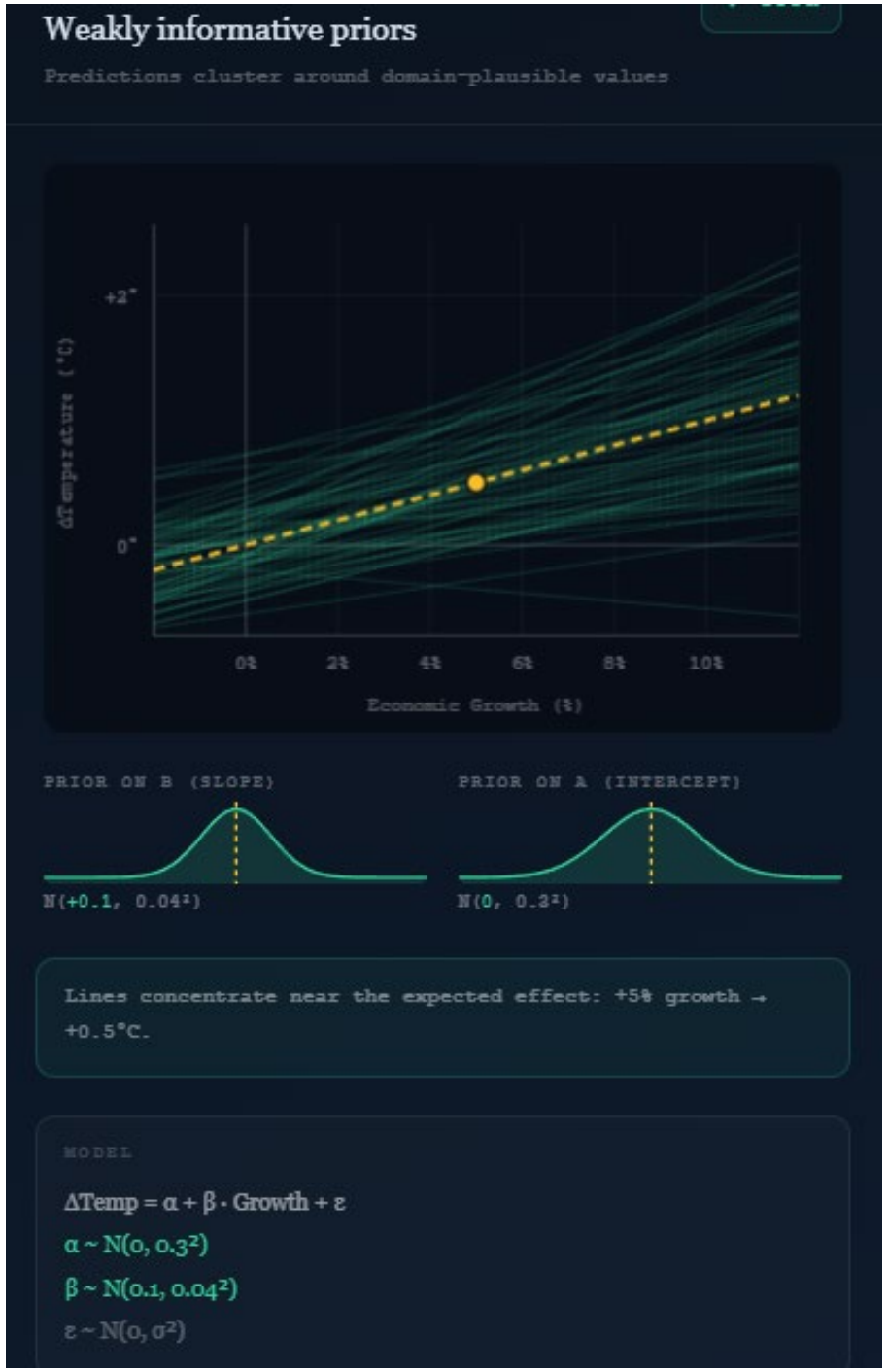


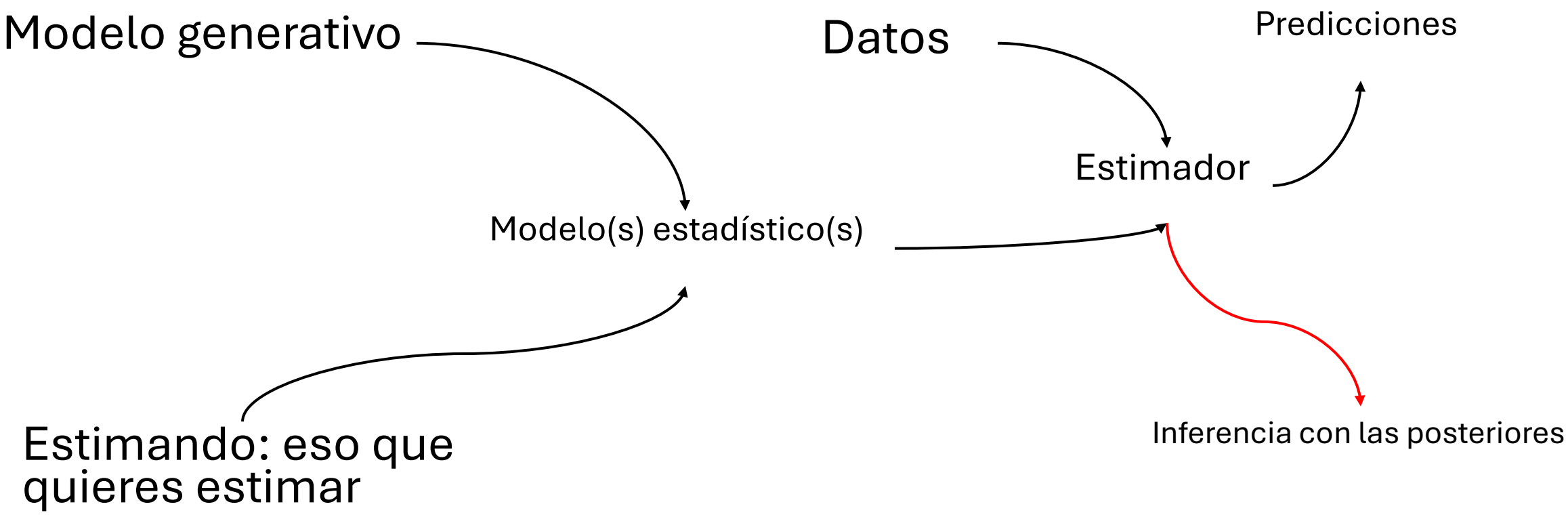
Efecto de priors demasiado vagos sobre las posibilidades que tiene el modelo generativo de crear valores razonables del parámetro de interés



Efecto de priors débilmente informativos sobre las posibilidades que tiene el modelo generativo de crear valores razonables del parámetro de interés

Efecto de priors menos débilmente informativos sobre las posibilidades que tiene el modelo generativo de crear valores razonables del parámetro de interés




Modelo generativo
Datos
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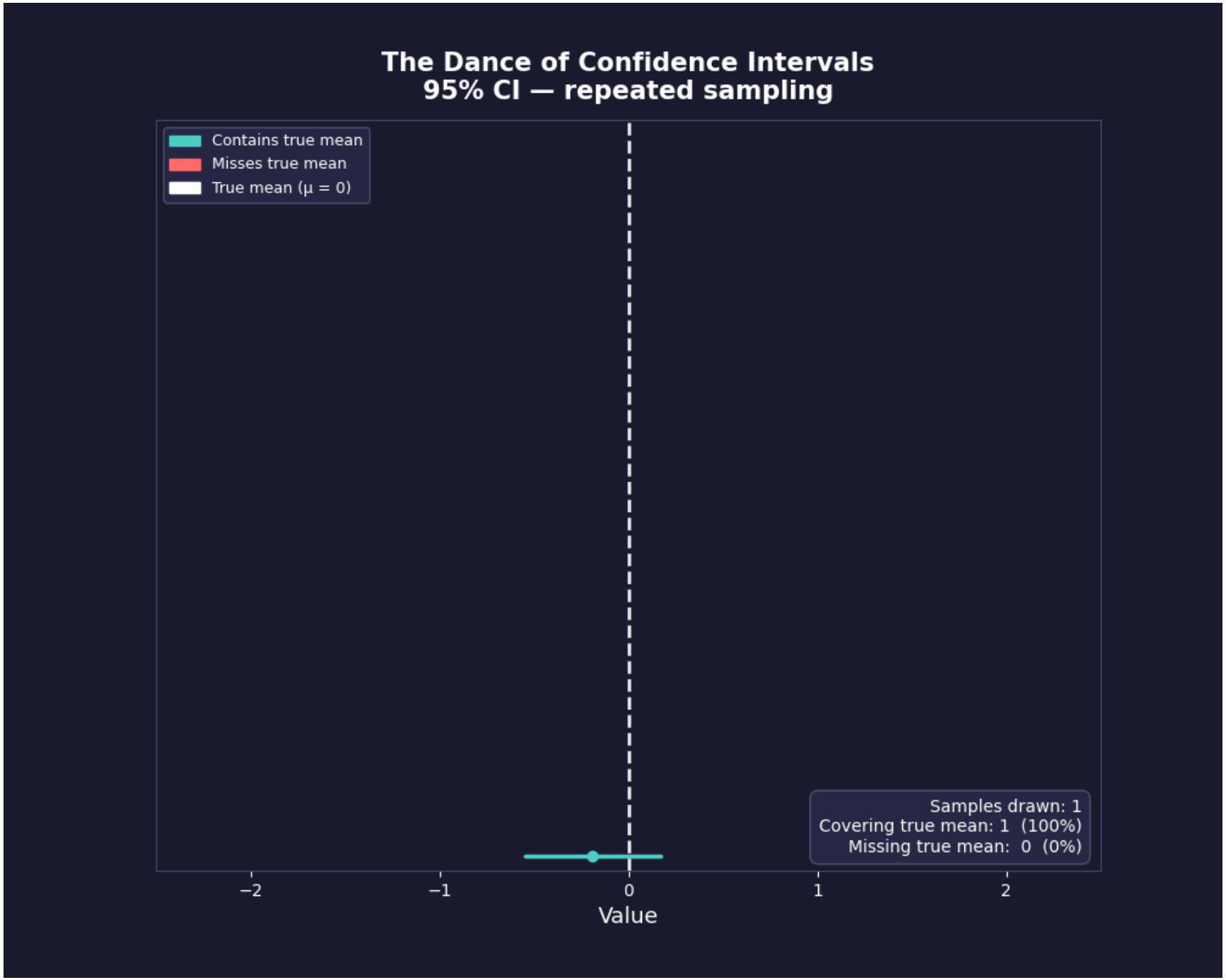
Estimando: eso que quieres estimar
Inferencia con las posteriores

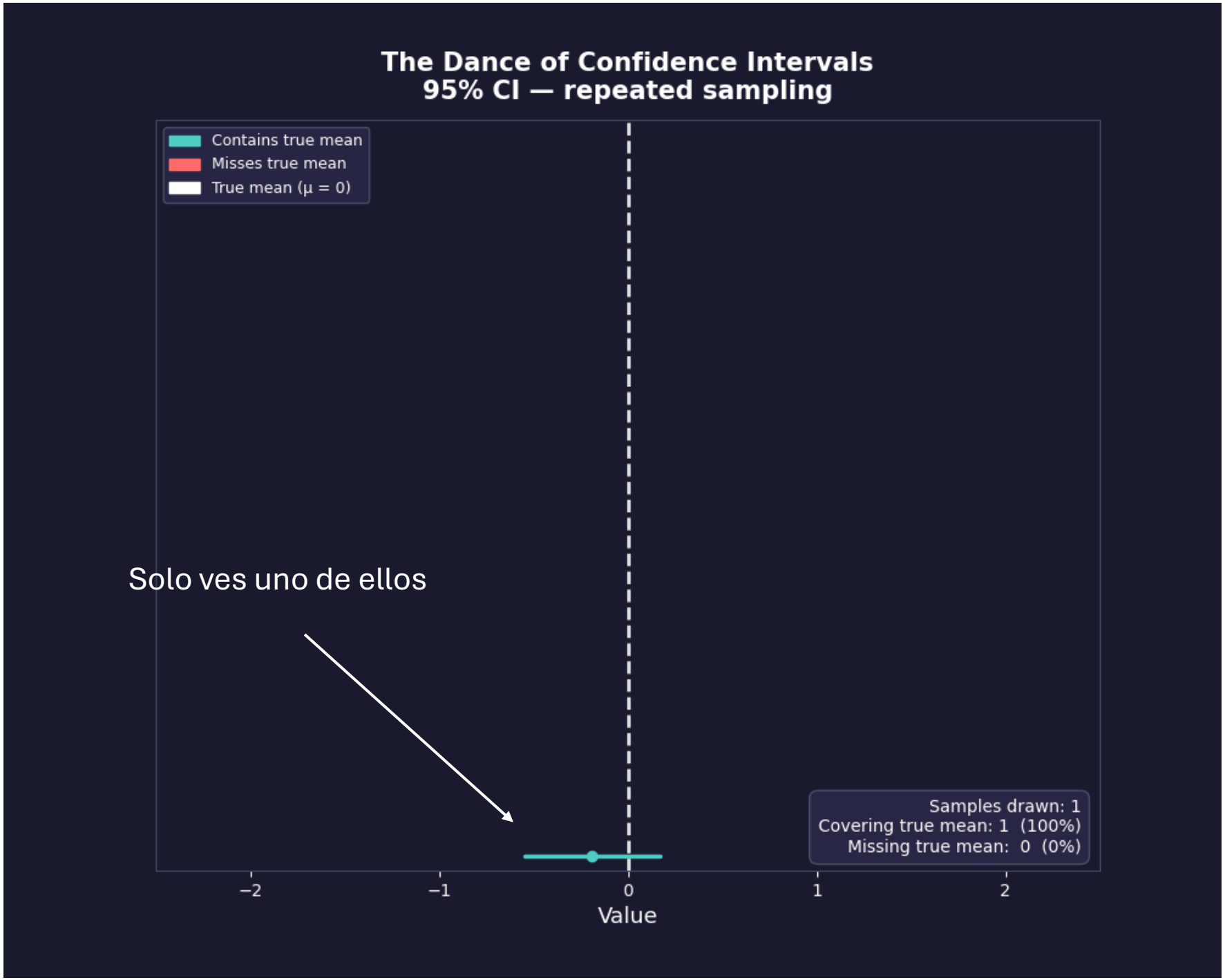
Medidas de incertidumbre

- Medidas explícitas y fáciles de interpretar de **qué tanto debemos de creer (incertidumbre),** dados los datos**,** que cierta hipótesis (la nula u otra, no importa) se sostiene.

- Medidas que emerjan en congruencia de un marco de inferencia probabilístico que atienda estos desafíos

¿Qué es el intervalo de confianza y qué me dice de la incertidumbre de mis estimaciones?




The Dance of Confidence Intervals
95% CI — repeated sampling
Contains true mean
Misses true mean
True mean (μ = 0)
Solo ves uno de ellos
Samples drawn: 1
Covering true mean: 1 (100%)
Missing true mean: 0 (0%)
−2
−1
0
1
2
Value

Inferencia estadística bajo incertidumbre

Conversación típica

Profesor frecuentista:

“95% de los intervalos de confianza contienen el valor verdadero del efecto del cambio en la tasa al ISR sobre la desigualdad”

Tesista:

“OK! Pero sólo veo un intervalo… ¿Entonces hay 95% de probabilidad de que el valor del stata esté en tal intervalo?”

Conversación típica

Profesor frecuentista (2):

“No, no. Los parámetros son fijos. El 95% es sobre el intervalo en cuestión.”

Tesista:

“OK! Pero 95/100 intervalos… ¿Entonces hay 95% de probabilidad de que el valor esté en tal intervalo?”

Conversación típica

Profesor frecuentista (3):

“No, no. Las propiedades de frecuencia nos dicen que 95% de los intervalos cachan al valor verdadero.”

Tesista:

“Eso lo explica todo. Por eso hay varios intervalos que no veo. ¿Entonces hay 95% de probabilidad de que el valor esté en tal intervalo?”

Conversación típica

Profesor frecuentista (4):

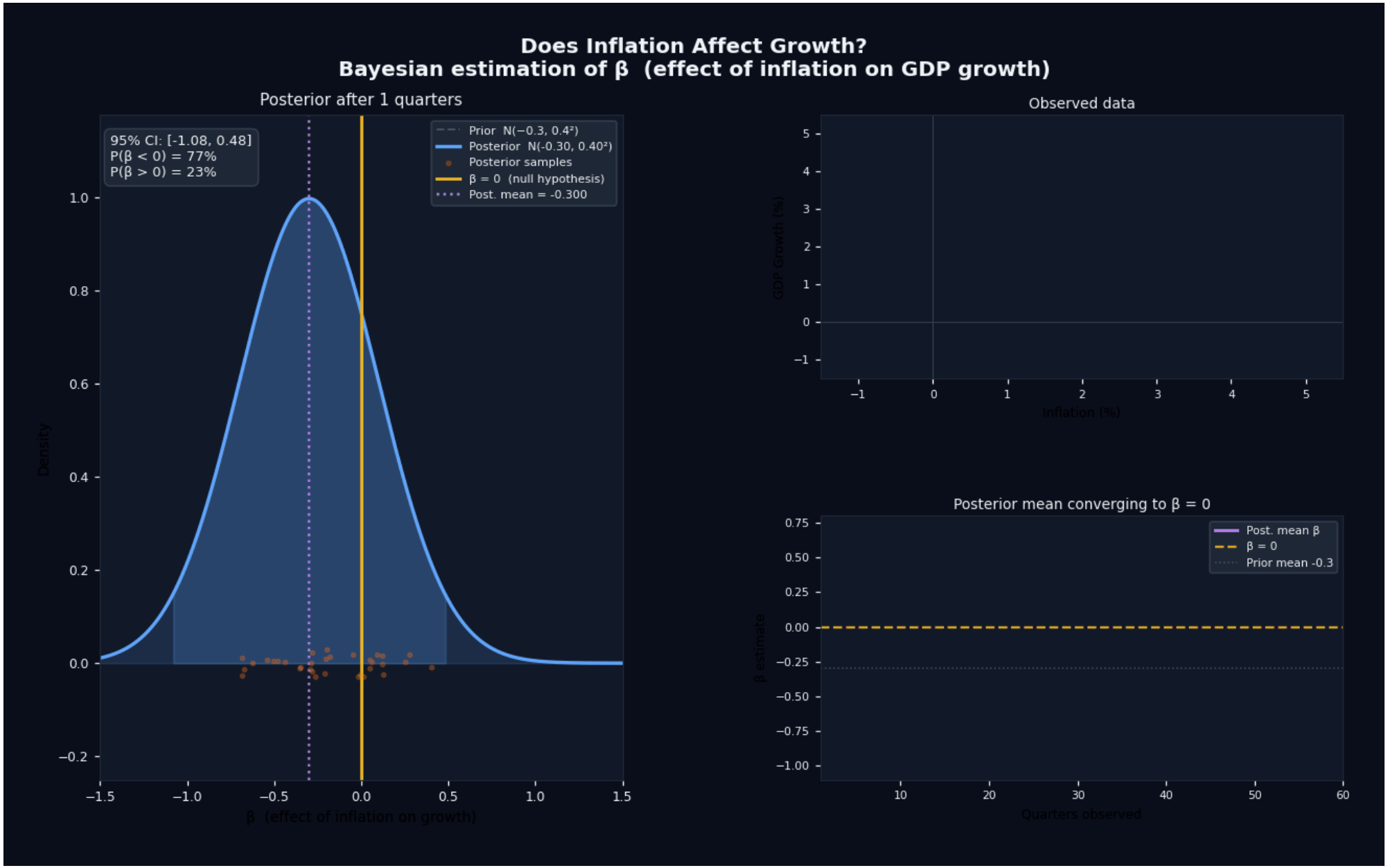
“Escribe la definición en el reporte”

Tesista:

“OK. Hay 95% de probabilidad de que el valor esté en tal intervalo”

El tesista es razonablemente necio: la inferencia es más razonable bajo distribuciones de parámetros

Intervalo creíble



Conversación atípica

Profesor bayesiano (5):

“Toma los descriptivos de la posterior”

Tesista:

“Condicional en el modelo, hay 95% de probabilidad de que le valor ronde entre -.03 y -.05 puntos del Theil”

Distintos tipos de inferencias sobre parámetros $p(\beta)$ de un modelo (ROPEs)

El efecto de terminar el posgrado en economía respecto a la licenciatura es:

$p(\theta|D)$: exactamente cero

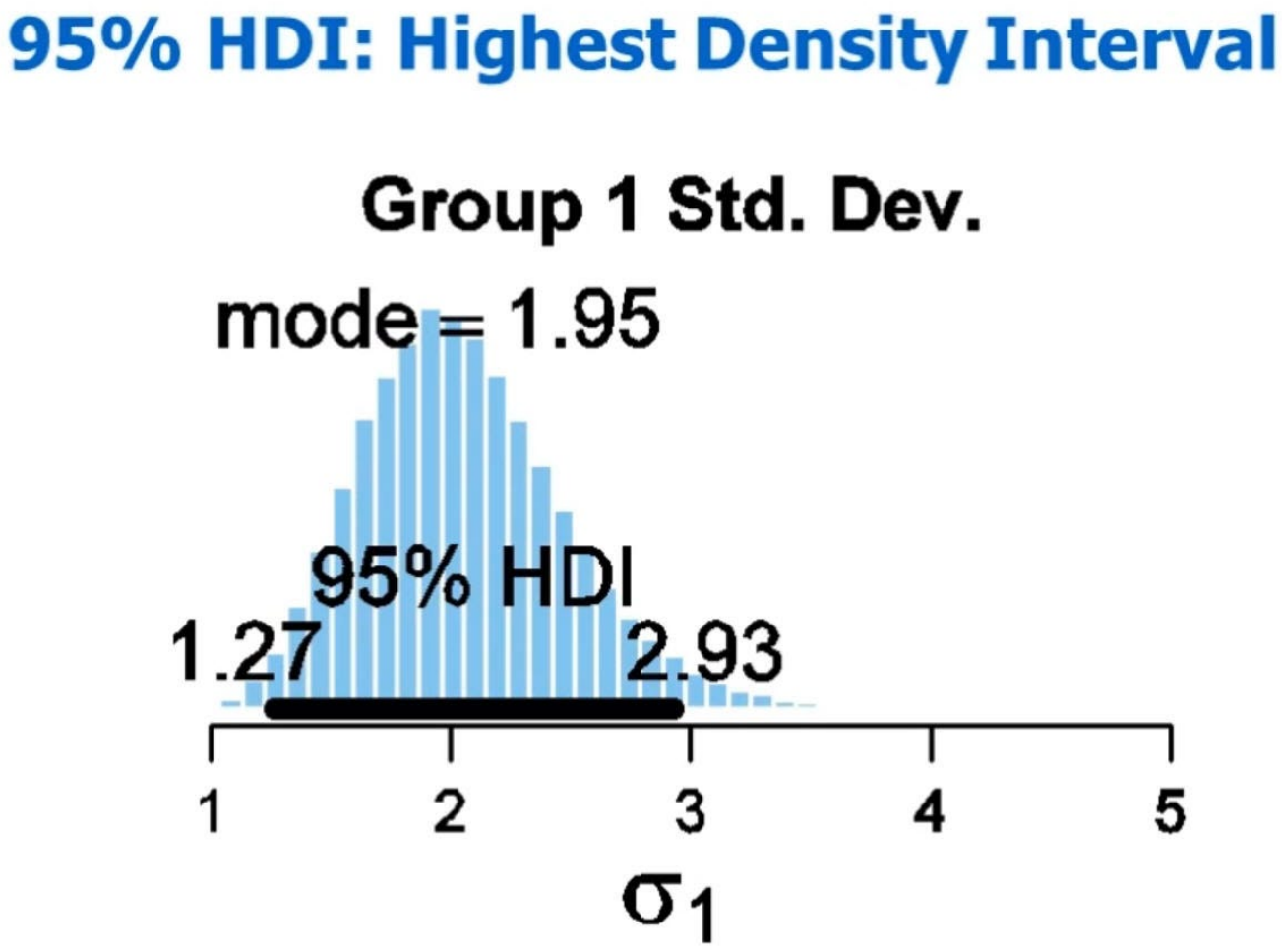
$p(\theta|D)$: positivo

$p(\theta|D)$: negativo

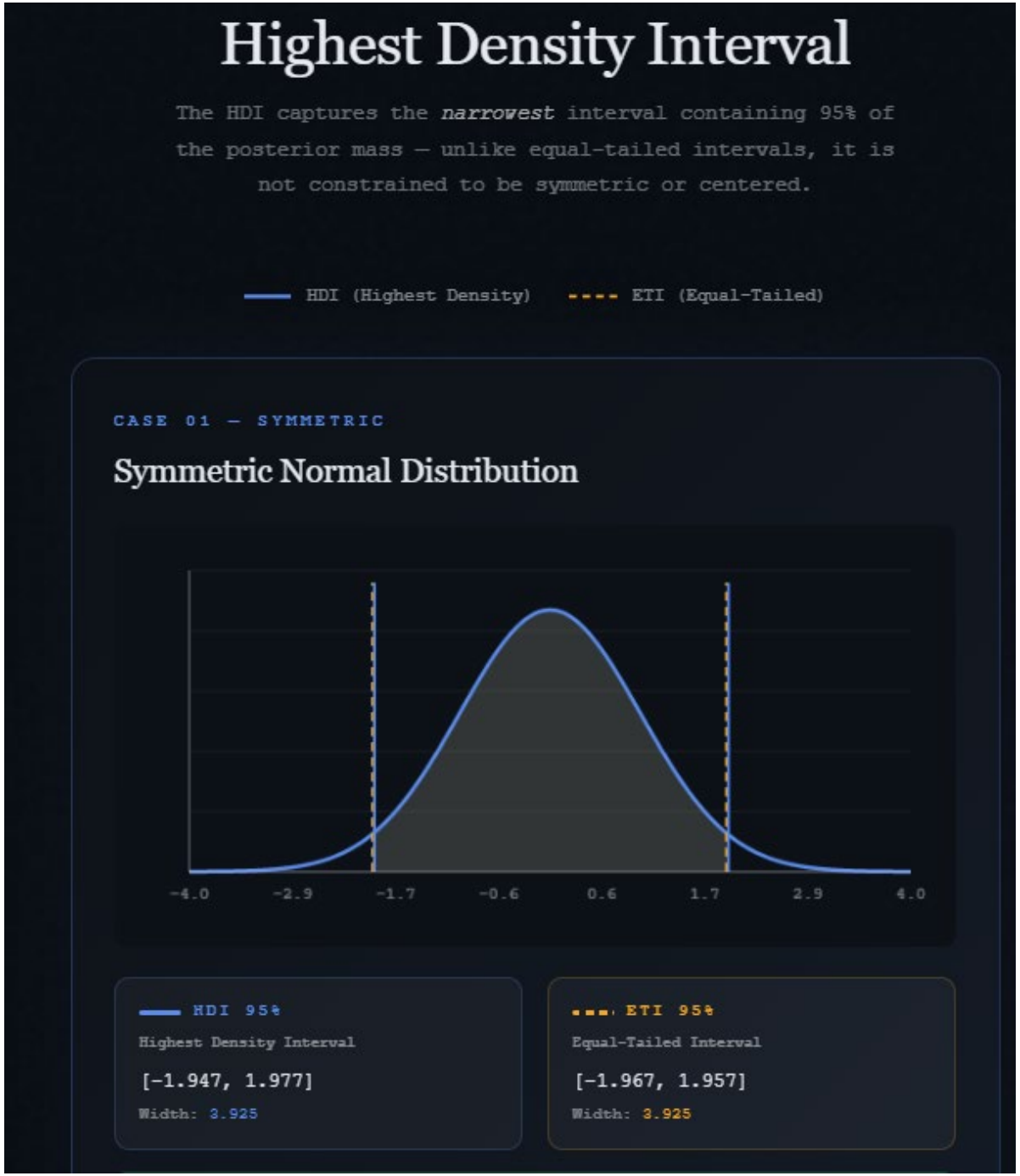
$p(\theta|D)$: positivo entre 9,000 y 12,000 pesos

$p(\theta|D)$: positivo entre 10,000 y 11,000 pesos

Intervalos de mayor densidad



Distribuciones simétricas



Distribuciones asimétricas

ROPE: Regiones de equivalencia

HDI inside ROPE

Reject H_0

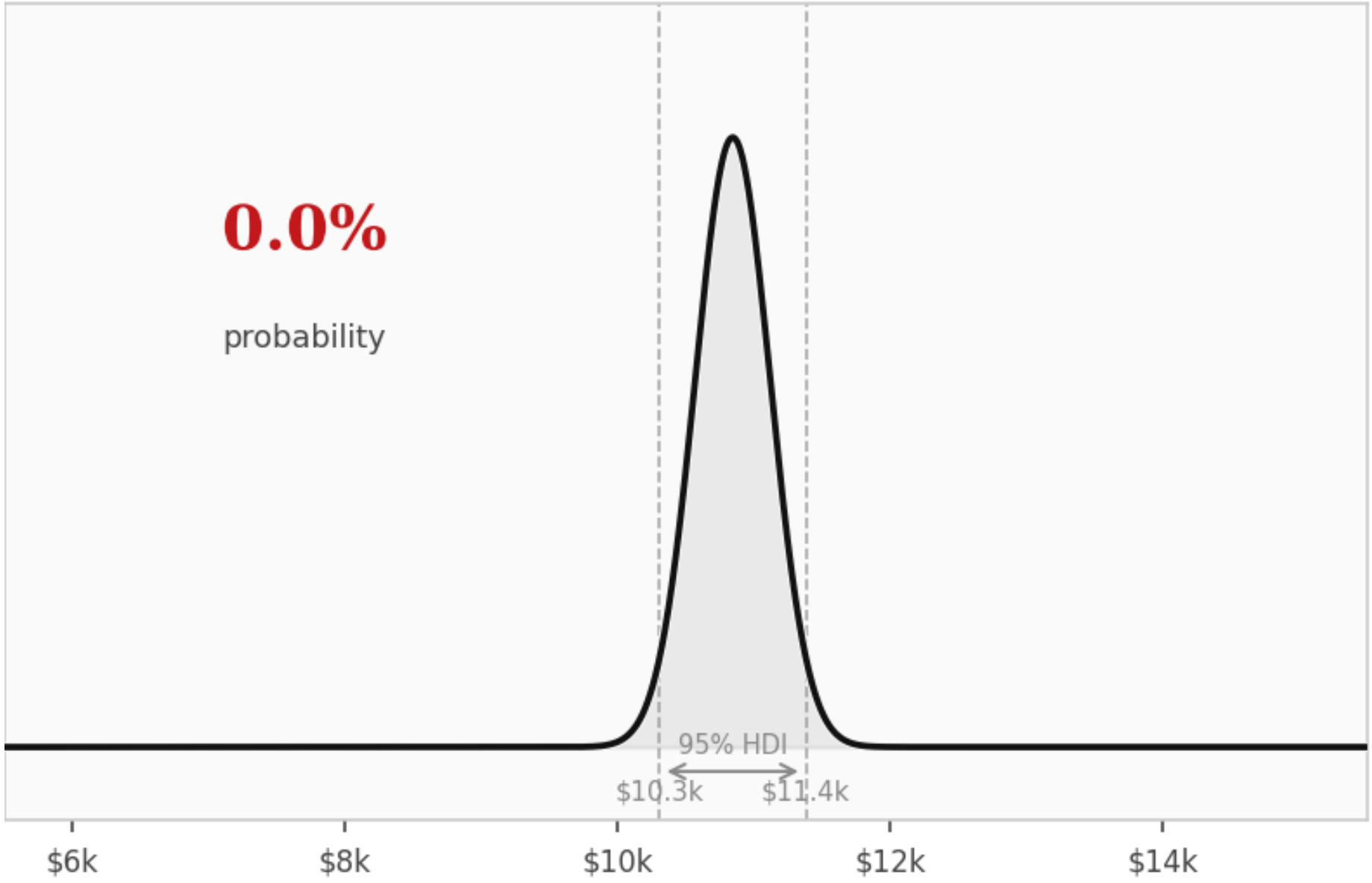
HDI falls entirely outside the ROPE

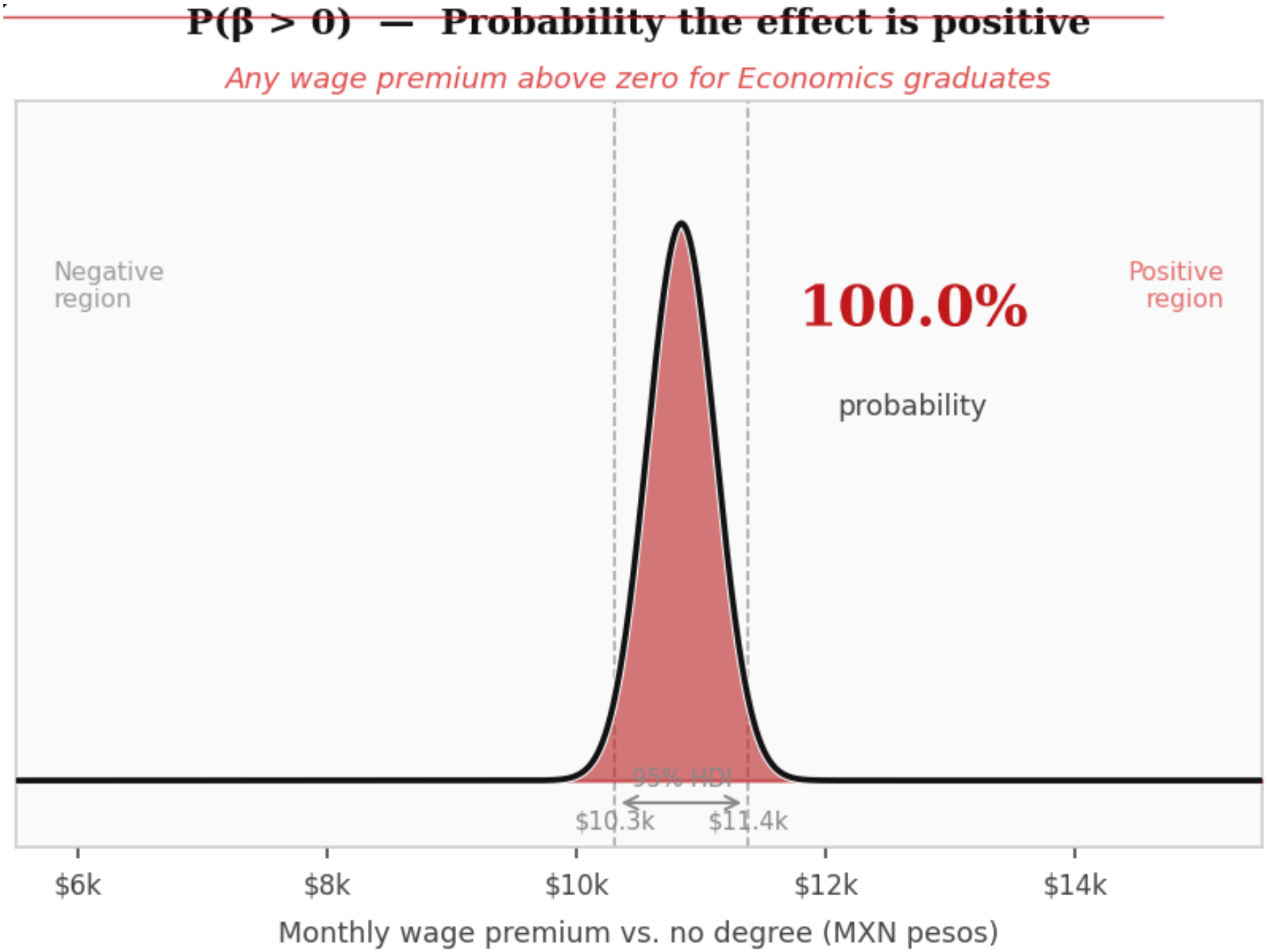
MASS IN ROPE

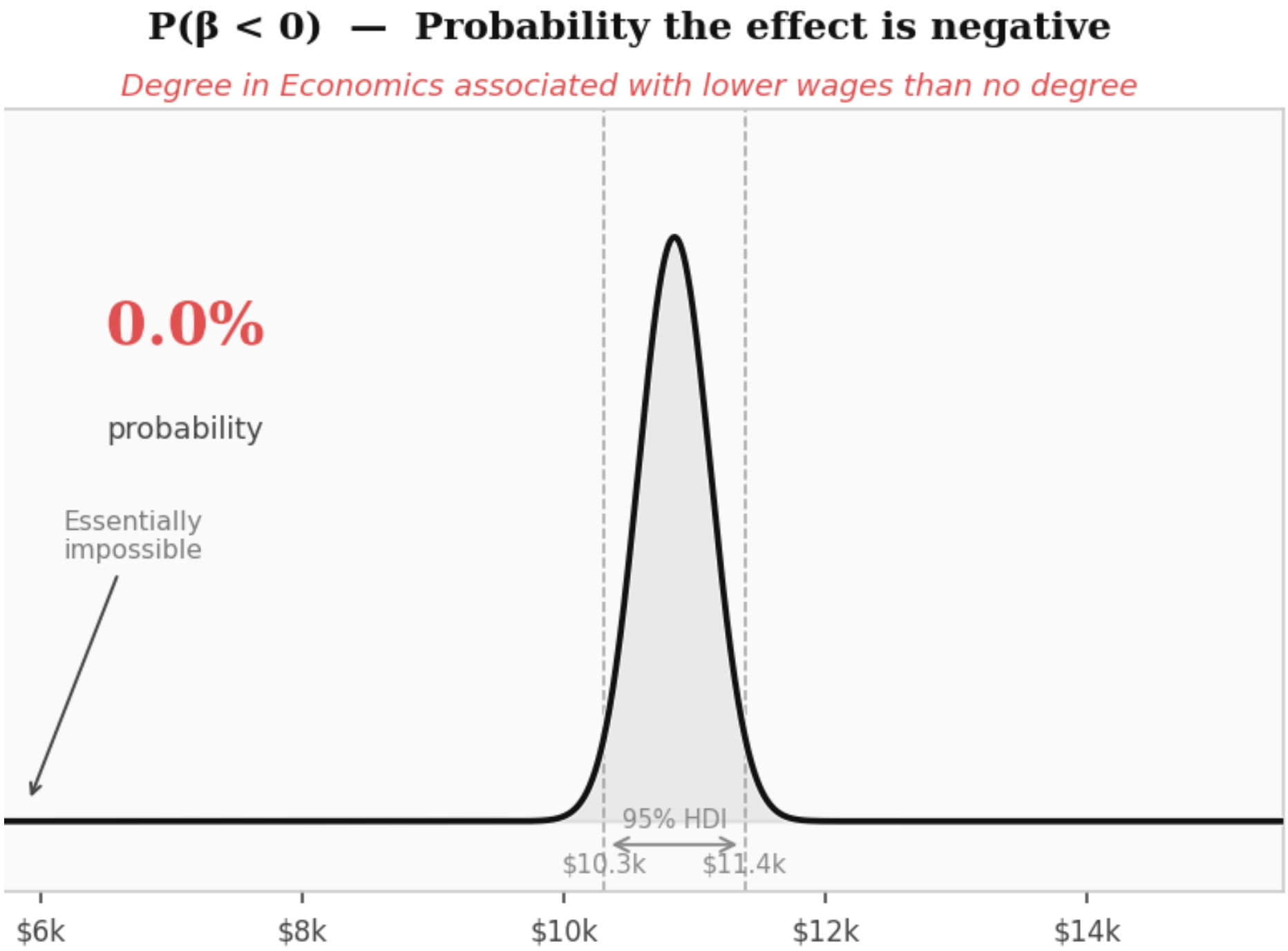
1.7%

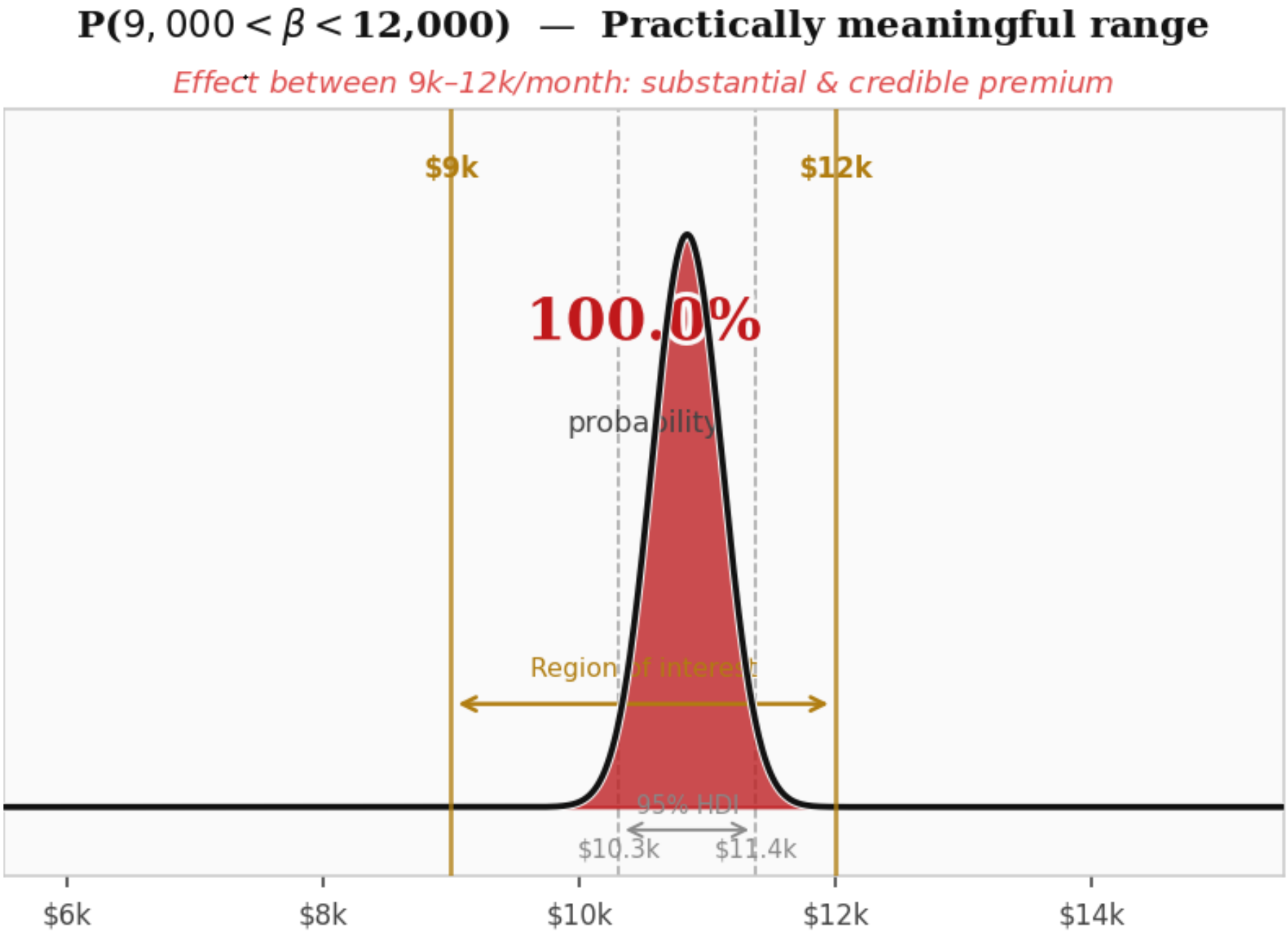
P(β = 0) — Probability the effect is exactly zero

Zero would mean a degree in Economics pays nothing extra



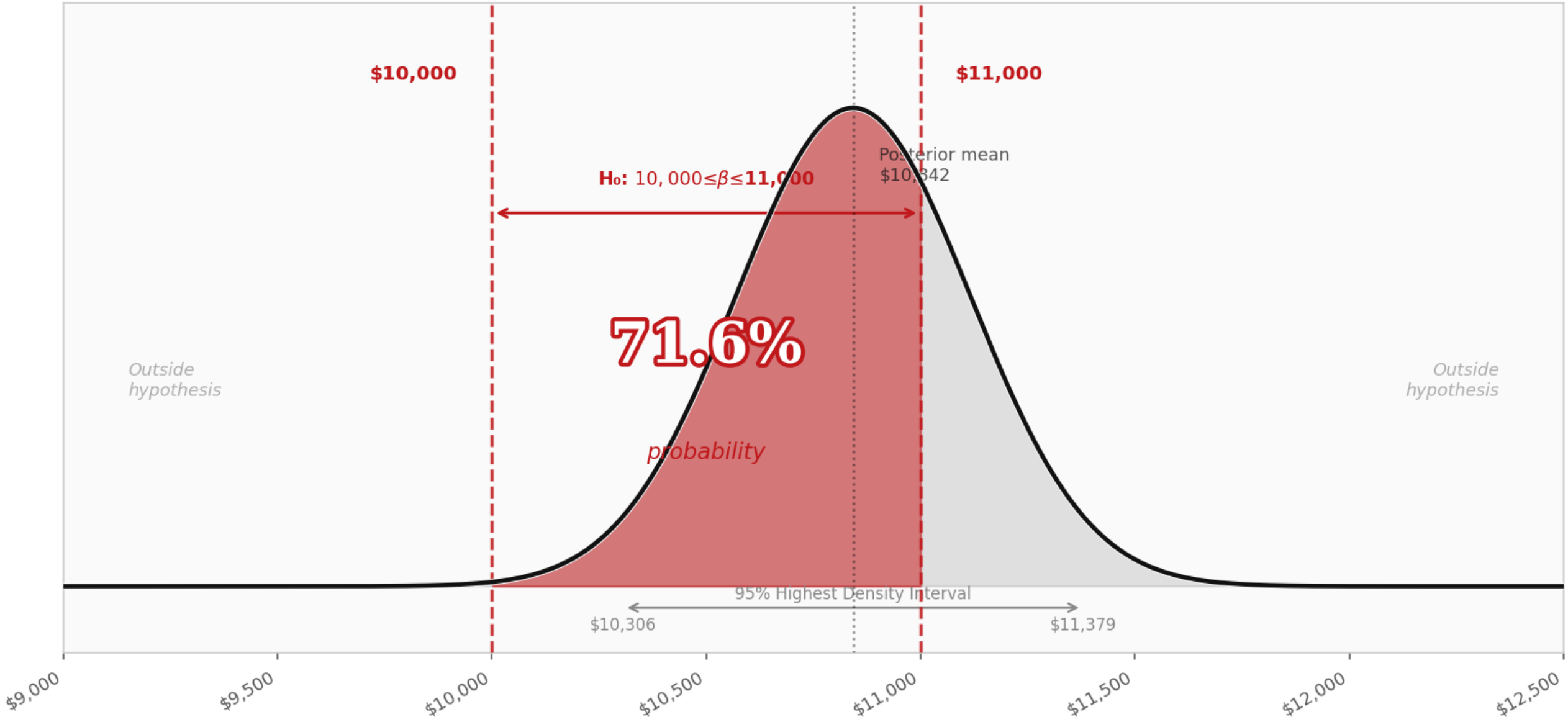

P(β > 0) — Probability the effect is positive
Any wage premium above zero for Economics graduates
Negative region
100.0%
probability
Positive region
95% HDI
$10.3k
$11.4k
$6k
$8k
$10k
$12k
$14k
Monthly wage premium vs. no degree (MXN pesos)


P(β < 0) — Probability the effect is negative
Degree in Economics associated with lower wages than no degree
0.0%
probability
Essentially
impossible
95% HDI
$10.3k
$11.4k
$6k
$8k
$10k
$12k
$14k


P(9,000 < β < 12,000) — Practically meaningful range
Effect between 9k–12k/month: substantial & credible premium
$9k
$12k
100.0%
probability
Region of interest
95% HDI
$10.3k
$11.4k
$6k
$8k
$10k
$12k
$14k

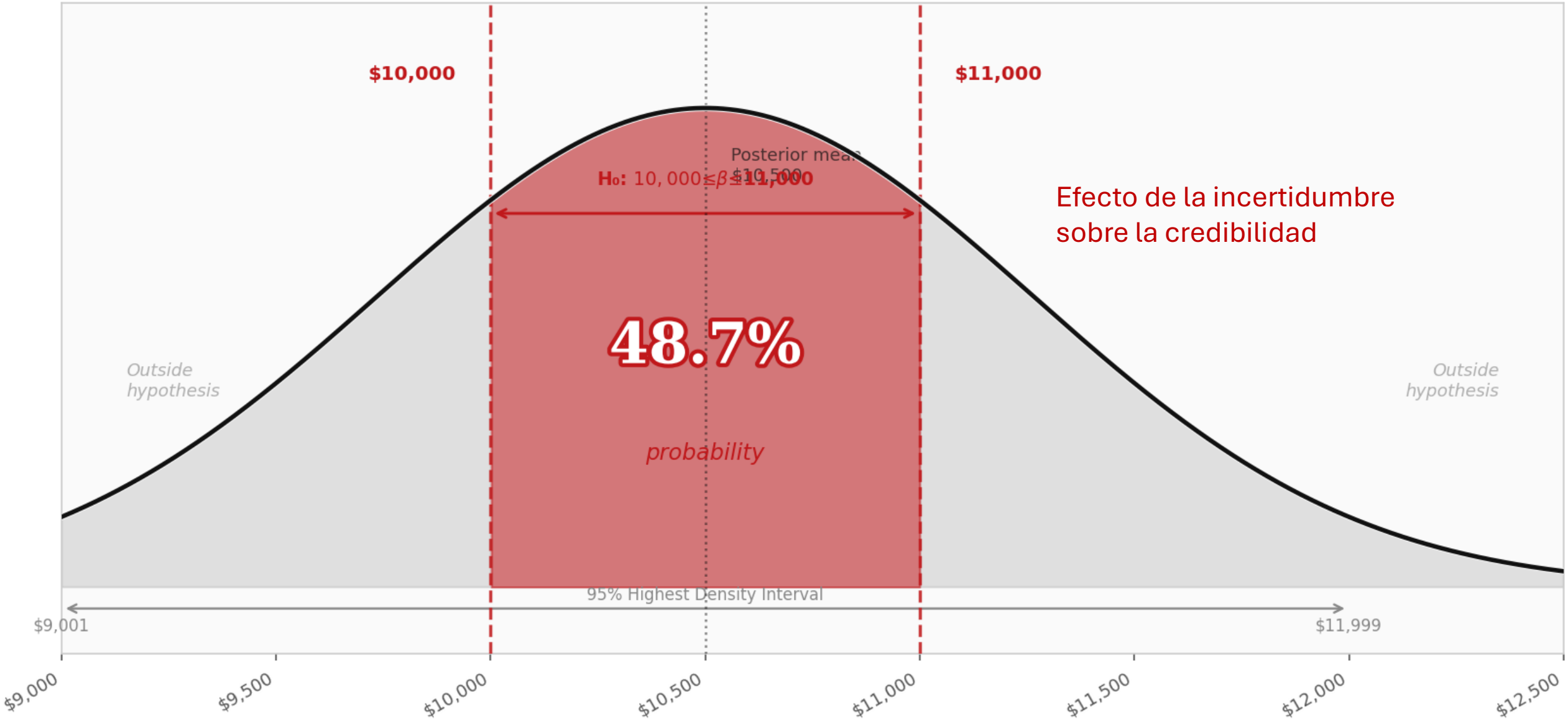
Returns to an Economics Degree

H_0: The wage premium is between 10,000and11,000 per month · Posterior N(10,842, 274) · n = 120

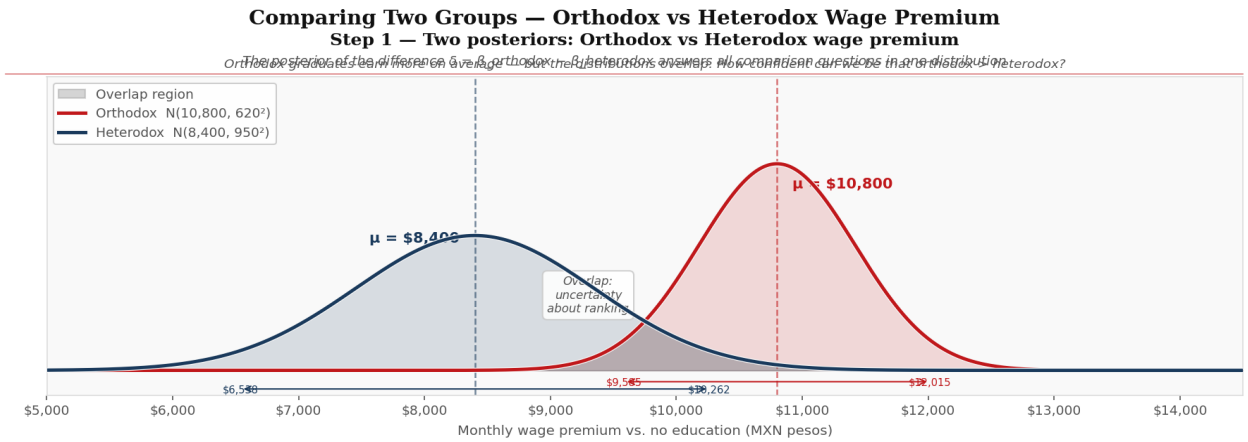


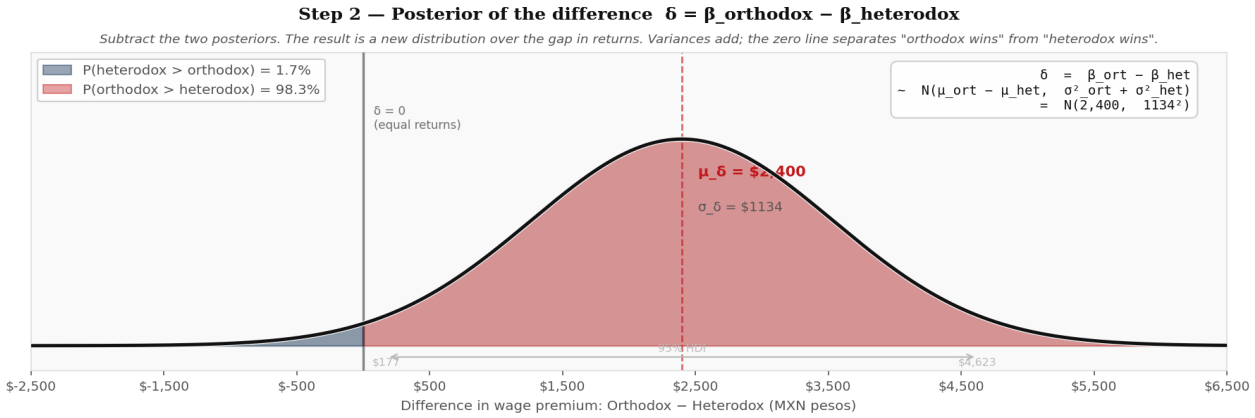
Returns to an Economics Degree

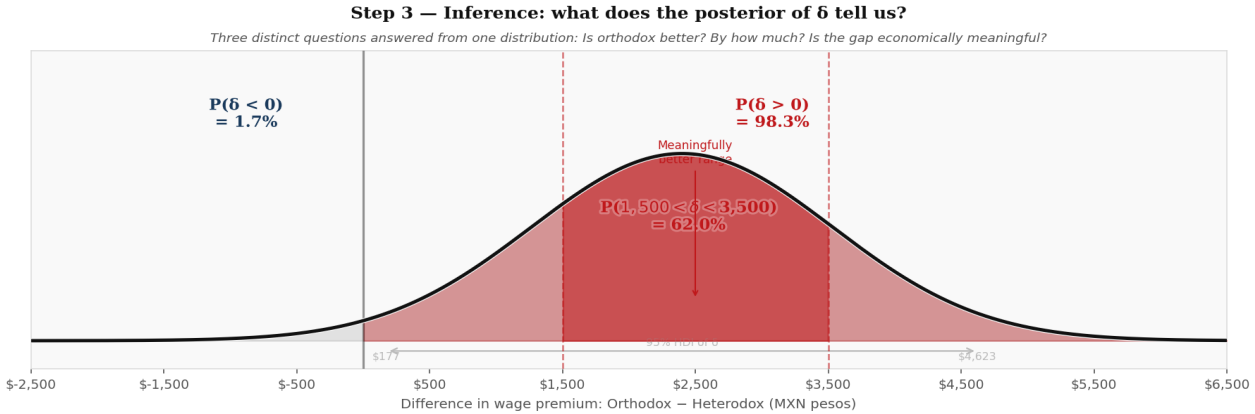
H_0: The wage premium is between 10,000 and 11,000 per month · Posterior N(10,500, 765) · n = 120



Comparación de parámetros / grupos







Posteriores (predicción y combinación) de parámetros globales

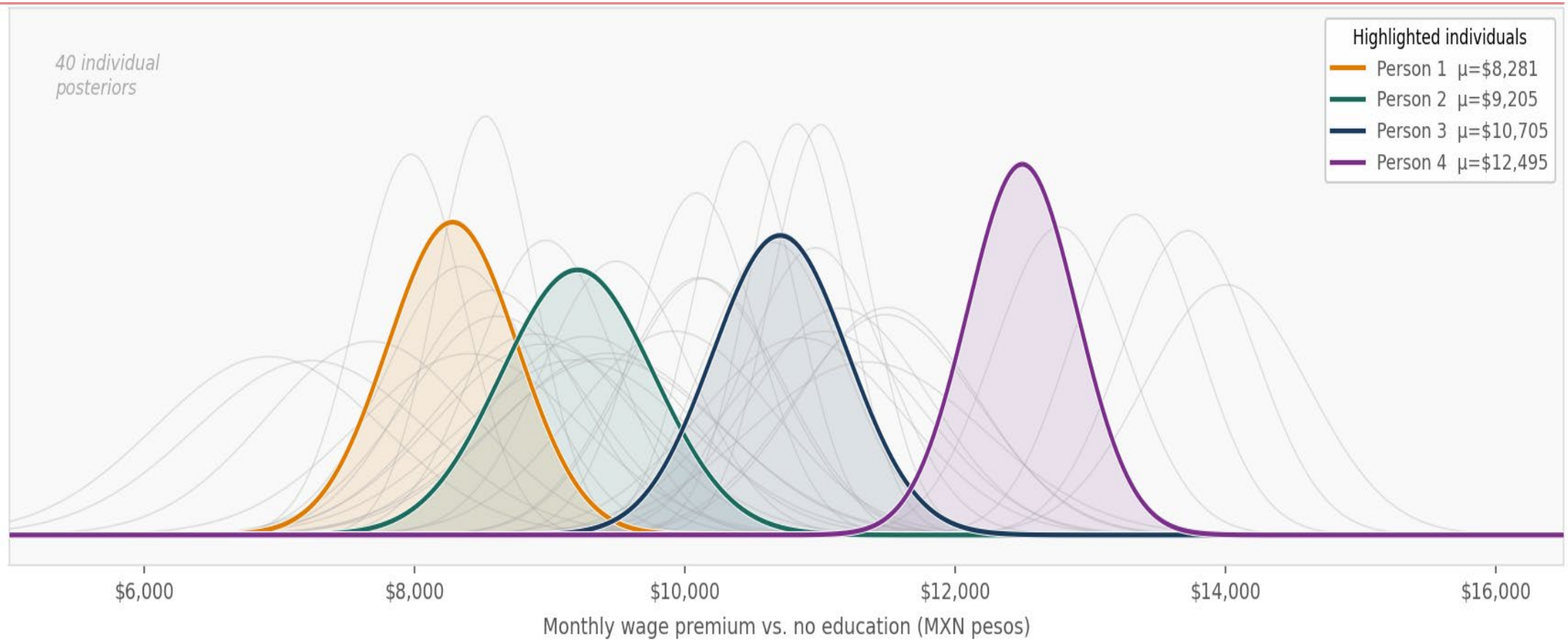
Imaginese que quisiéramos calcular el salario promedio de 40 egresados $p(\theta|D)$.

Un buen modelo generativo tendría que condicionar por varios factores: edad, sexo, año de graduación, campo de conocimiento, etc.

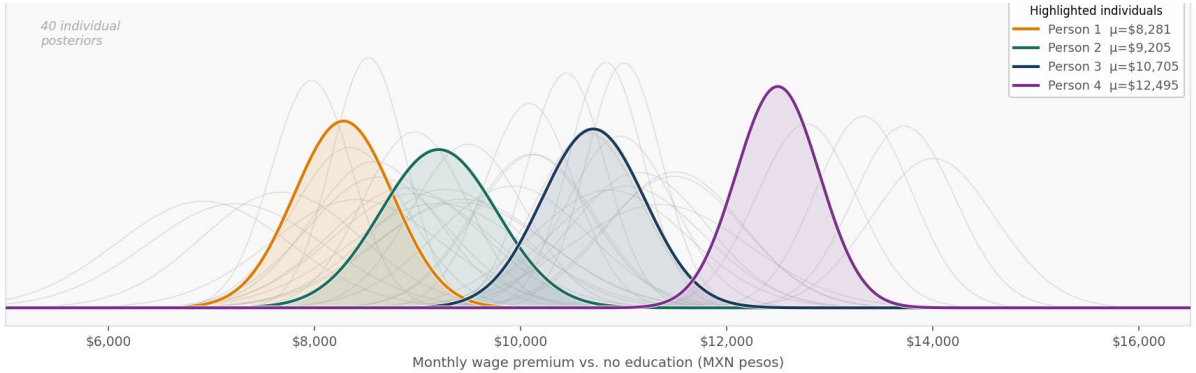
Bajo tal modelo, obtendríamos:

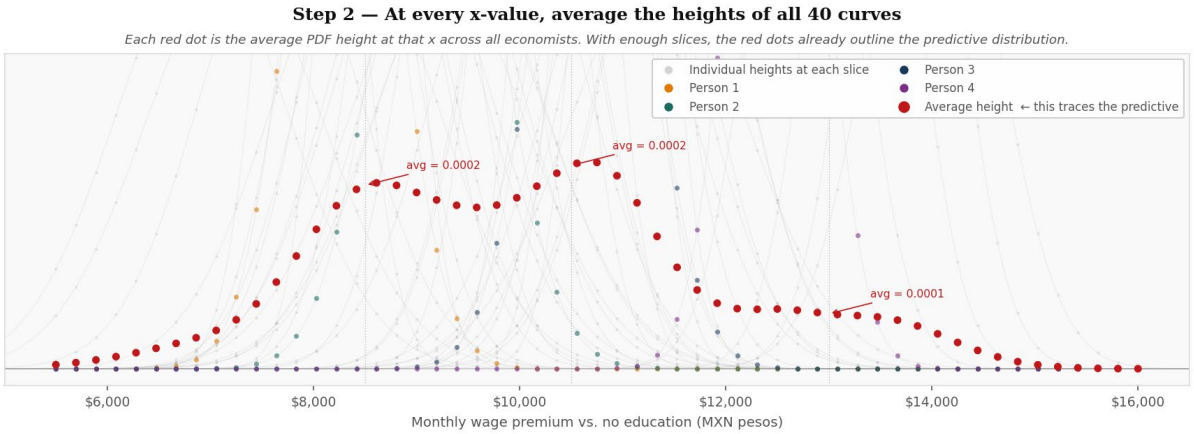
- Las posteriores de los parámetros del modelo (láminas anteriores)

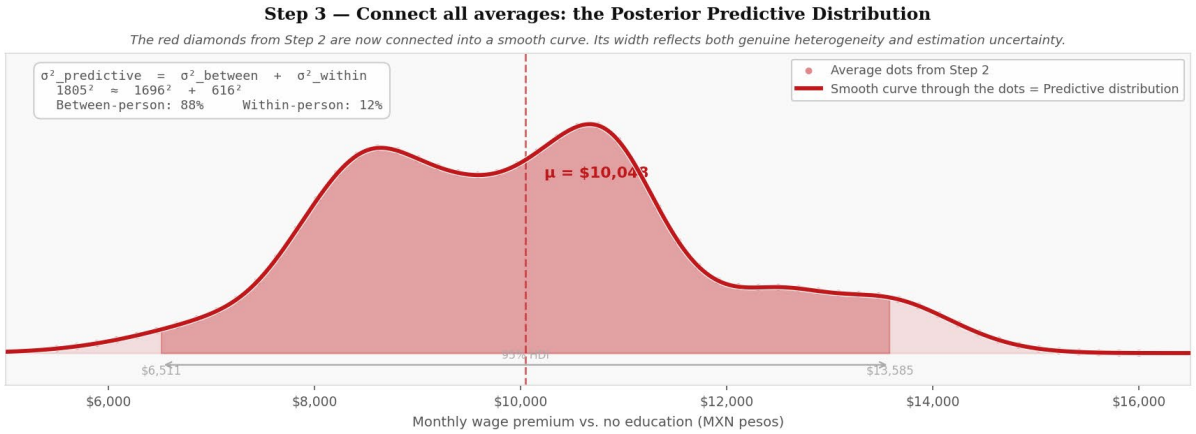
Los parámetros generan la predicción del salario esperado de cada estudiante $p(\theta_i|D)$



La media de las posteriores permite estimar la media de toda la población junto con su incertidumbre







De vuelta al papel de los priors: cómputo (próxima clase)

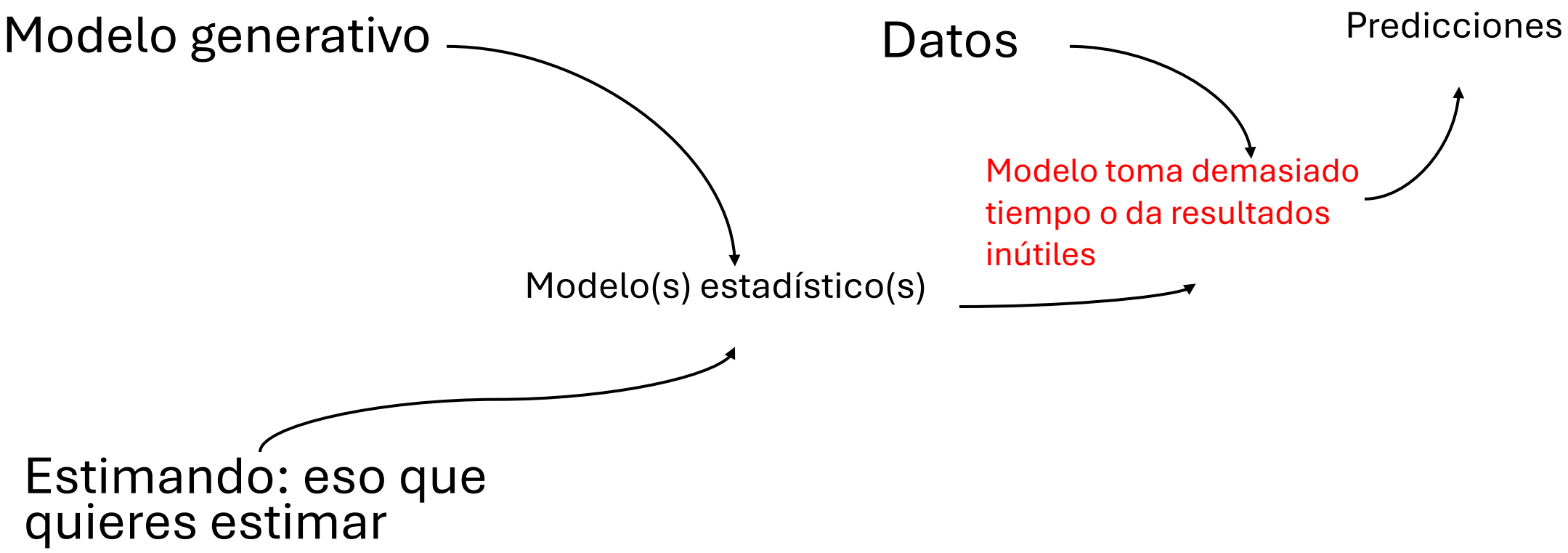


Table 5.5: Applying Bayes' rule to data and parameters. When conditionalizing on row value D, the conditional probability $p(\theta|D)$ is the cell probability $p(D,\theta)$ divided by the marginal probability $p(D)$. When these probabilities are algebraically re-expressed as shown in the table, this is Bayes' rule. This table is merely Table 5.1 with its rows and columns re-named. Copyright © Kruschke, J. K. (2014). *Doing Bayesian Data Analysis: A Tutorial with R, JAGS, and Stan. 2nd Edition.* Academic Press / Elsevier.

Priors

Data	**Model Parameter**			Marginal
	...	θ value	...	
$\vdots$		$\vdots$		$\vdots$
D value	...	$p(D,\theta)$ $= p(D\|\theta)\,p(\theta)$	...	$p(D)$ $= \sum_{\theta^*} p(D\|\theta^*)\,p(\theta^*)$
$\vdots$		$\vdots$		$\vdots$
Marginal	...	$p(\theta)$	...	

La probabilidad de cada parámetros /
La probabilidad de cada D (renglón)

Para el caso continuo es la integral

DATOS ACTIVOS $x_1 \ldots x_4$

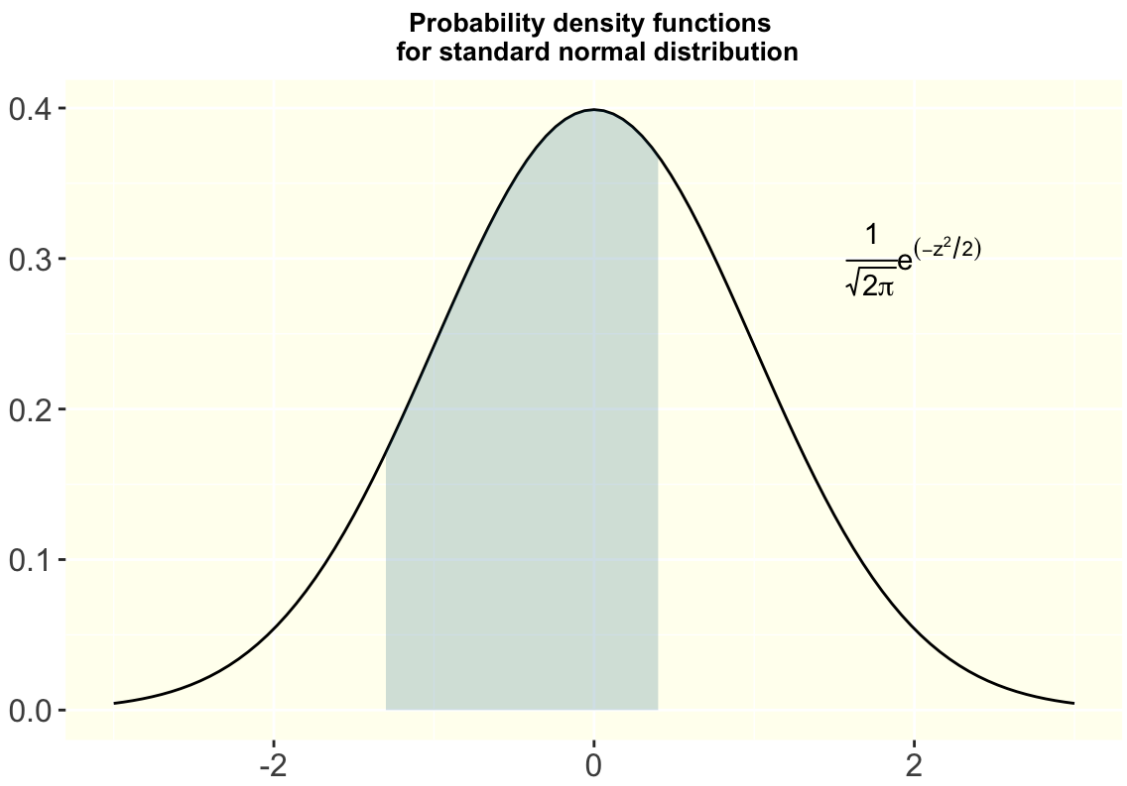
$x_1 = +0.52$ | $x_2 = -0.31$ | $x_3 = +1.07$ | $x_4 = -0.88$ | $x_5 = +0.23$ | $x_6 = -0.64$ | $x_7 = +0.79$ | $x_8 = -0.15$

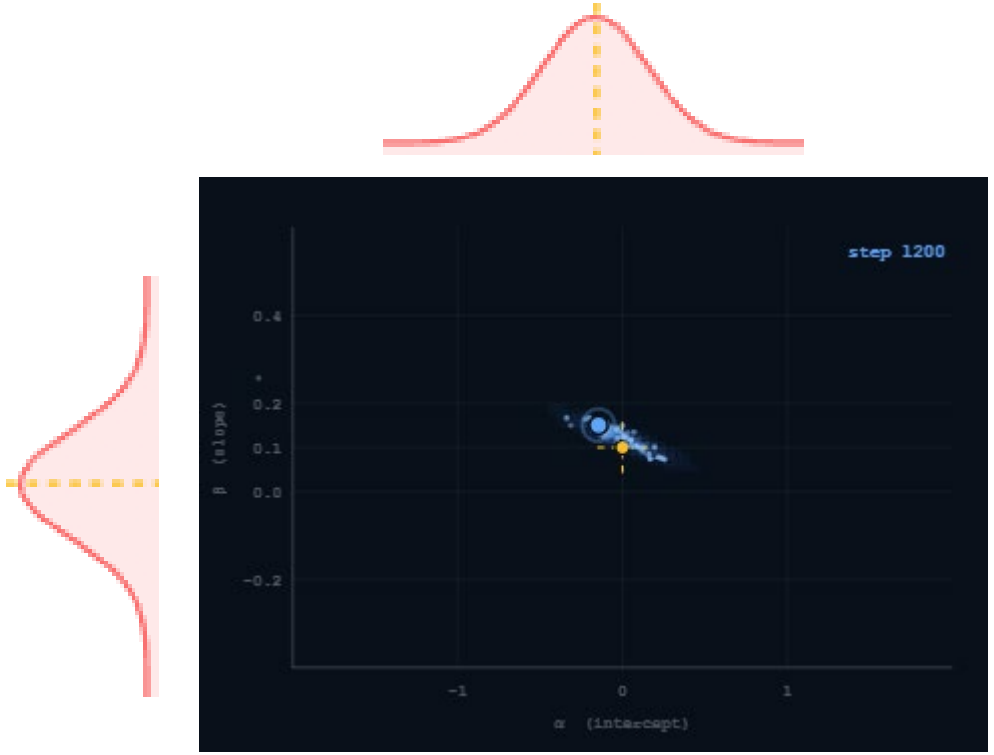
θ candidato	Descripción	p(x1|θ) +0.52	p(x2|θ) −0.31	p(x3|θ) +1.07	p(x4|θ) −0.88	$p(x_1 \ldots x_4 \mid \theta)$ verosimilitud conjunta	p(θ) prior N(0,2²)	p(θ|x) ∝ posterior (sin norm.)
-3.00	Far from data	0.001	0.011	0.000	0.042	3.71e-11	0.0648	2.40e-12
-2.00	Unlikely region	0.017	0.096	0.004	0.213	0.000001	0.1210	1.47e-7
-1.00	Below typical set	0.126	0.314	0.047	0.396	0.000733	0.1760	0.000129
-0.50	Edge of typical set	0.237	0.392	0.116	0.371	0.004011	0.1933	0.000776
0.00 ★	True value / MLE	0.348	0.380	0.225	0.271	0.008078	0.1995	0.001611
0.12	Near true value	0.368	0.364	0.254	0.242	0.008234	0.1991	0.001640 ▲
0.50	Edge of typical set	0.399	0.287	0.339	0.154	0.005984	0.1933	0.001157
1.00	Above typical set	0.356	0.169	0.398	0.068	0.001631	0.1760	0.000287
2.00	Unlikely region	0.133	0.028	0.259	0.006	0.000006	0.1210	7.30e-7

Los priors ayudan a delimitar el espacio de solución:

Eficiencia para el cómputo

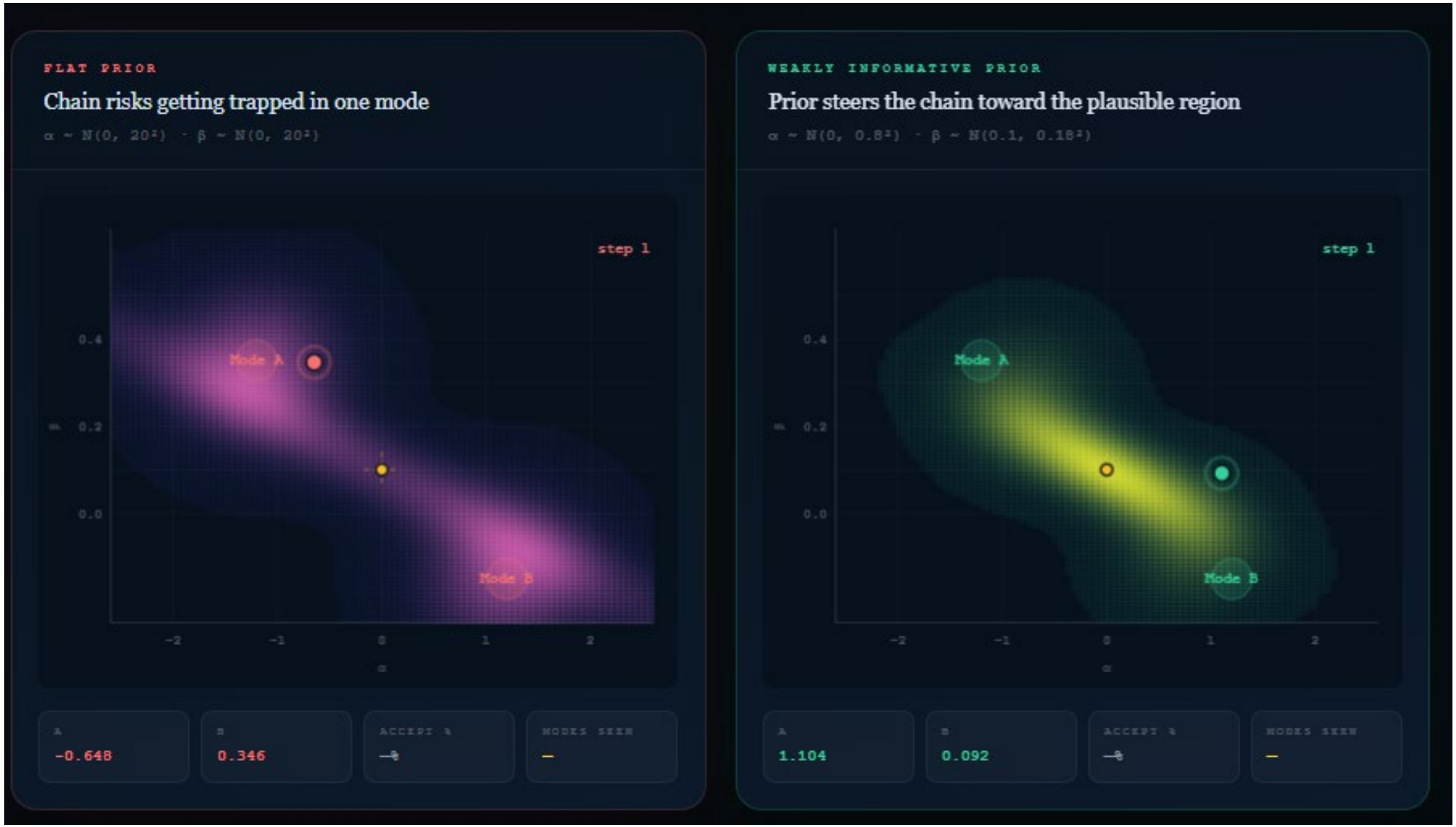
Modelos de dos parámetros





necesitamos integrar sobre todas las posibilidades de los valores posible de los parámetros

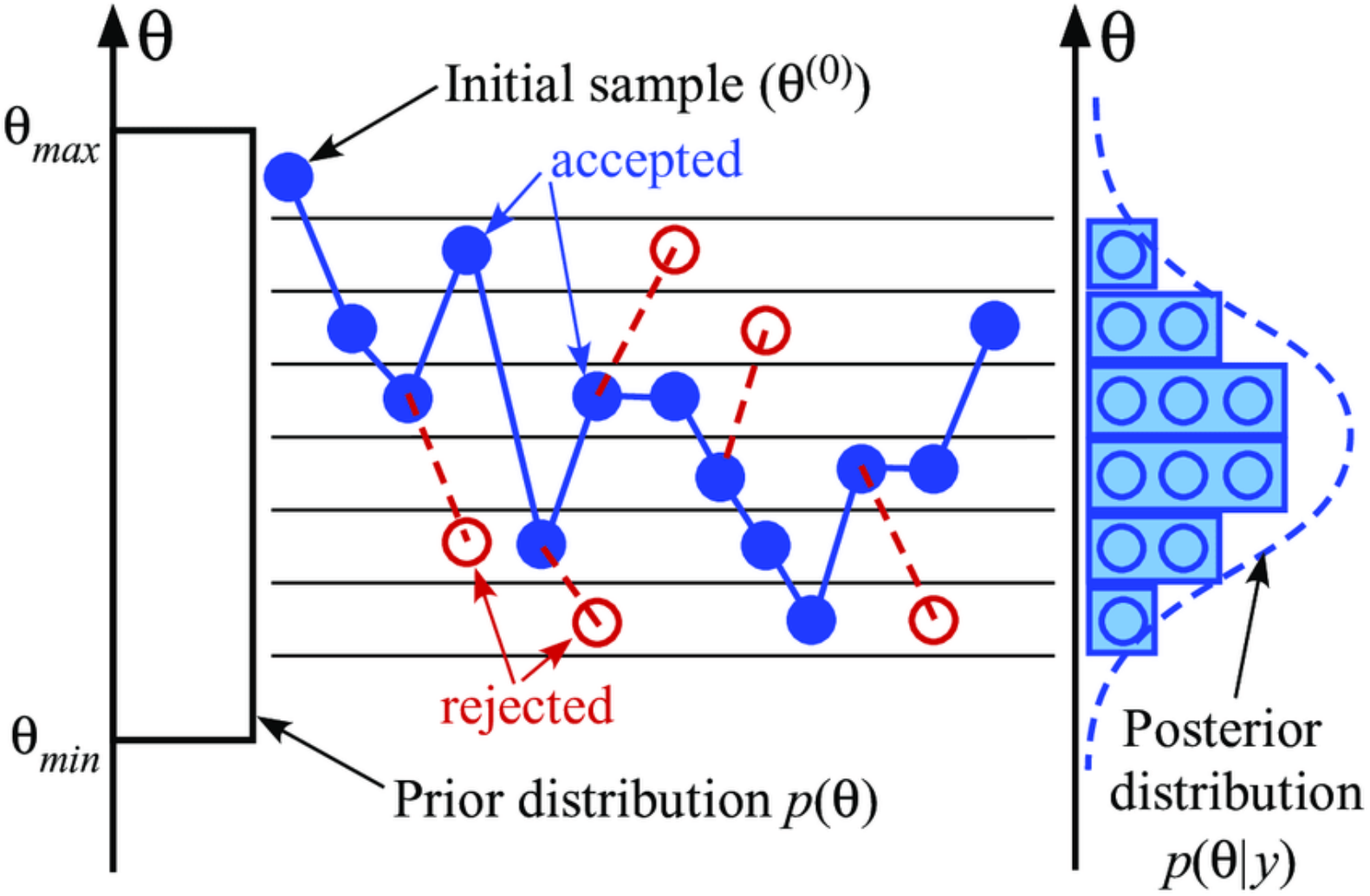
Priors para regularizar la estimación



Próxima clase

McElreath. Capítulo 8

Kruschke. Capítulo 7



Referencias

- Gelman, A., Simpson, D., & Betancourt, M. (2017). The prior can often only be understood in the context of the likelihood. *Entropy*, *19*(10), 555.
- Van Dongen, S. (2006). Prior specification in Bayesian statistics: three cautionary tales. *Journal of theoretical biology*, *242*(1), 90-100.